

Huawei AI Certification Training

HCIA-AI

Machine Learning Lab Guide

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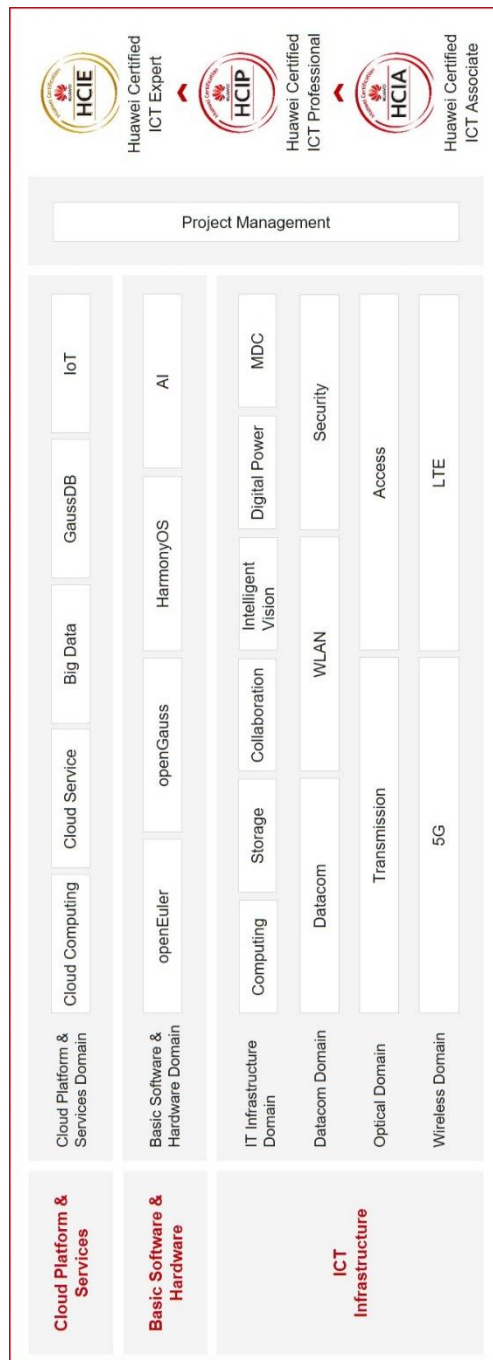
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About This Document

Overview

This document applies to candidates who are preparing for the HCIA-AI exam and others who want to learn basic machine learning knowledge.

Description

This lab guide comprises the following five experiments:

- Experiment 1 - Linear regression. The Python tool package scikit-learn is used to implement a simple linear regression algorithm.
- Experiment 2 - Linear regression expansion. The Python basic tool package numpy is used to implement algorithms such as linear regression and gradient descent from scratch.
- Experiment 3 - Logistic regression. The logistic regression algorithm in the tool package is used to implement simple classification.
- Experiment 4 - Decision tree. Trainees will construct a decision tree to predict the weather, and visualize the decision tree.
- Experiment 5 - K-means clustering algorithm.

Background Knowledge Required

This course is for Huawei's professional certification. Trainees are expected to meet the following requirements:

- Have basic knowledge of Python programming.
- Have basic math skills, including linear algebra and probability theory.

Lab Environment Overview

The lab environment is Python 3.7. For details about how to set up the environment, see the Environment Setup Guide.

Datasets used in this lab can be obtained at <https://certification-data.obs.cn-north-4.myhuaweicloud.com/ENG/HCIA-AI/V4.0/Dataset.zip>

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1 Implementation of Common Machine Learning Algorithms

1.1 Introduction

1.1.1 About This Lab

This lab introduces common machine learning algorithms to help you better understand their functions and usages. Specifically, it will explain how to build a linear regression algorithm from scratch and implement the decision tree and K-means clustering based on scikit-learn.

1.1.2 Objectives

- Build a linear regression algorithm from scratch.
- Master the use of classification and regression algorithms.
- Master the use of the V clustering algorithm.
- Master the implementation process of machine learning algorithms.

1.2 Code Implementation

1.2.1 Linear Regression

Step 1 Import dependent packages.

Input:

```
from sklearn.linear_model import LinearRegression # Import the linear regression model
import matplotlib.pyplot as plt # The plotting library
import numpy as np
```

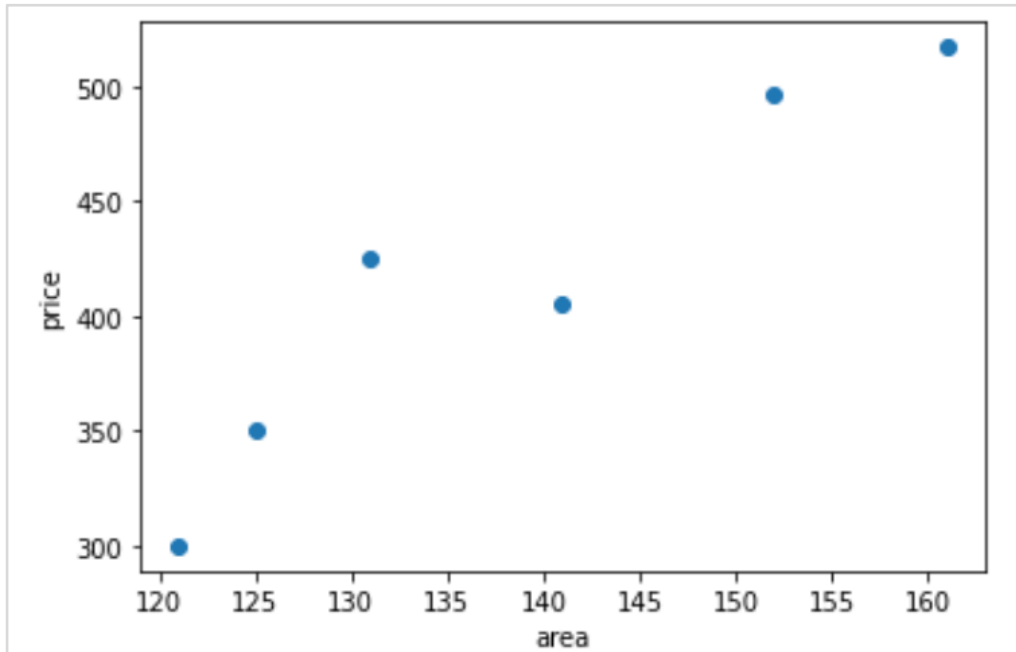
Step 2 Build and visualize a house price dataset.

Input:

```
x = np.array([121, 125, 131, 141, 152, 161]).reshape(-1,1) # x denotes the house area as a feature.
y = np.array([300, 350, 425, 405, 496, 517]) # y denotes the house price.
plt.scatter(x,y)
plt.xlabel("area") # X axis indicates the area.
plt.ylabel("price") # Y axis indicates the price.
```

```
plt.show()
```

Output:



Step 3 Train the model.

Input:

```
lr = LinearRegression() # Encapsulate the linear regression model into an object.
lr.fit(x,y) # Train the model on the dataset.
```

Step 4 Visualize the model.

Input:

```
w = lr.coef_# Slope of the model
b = lr.intercept_# Intercept of the model
print('Slope: ',w)
print('Intercept: ',b)
```

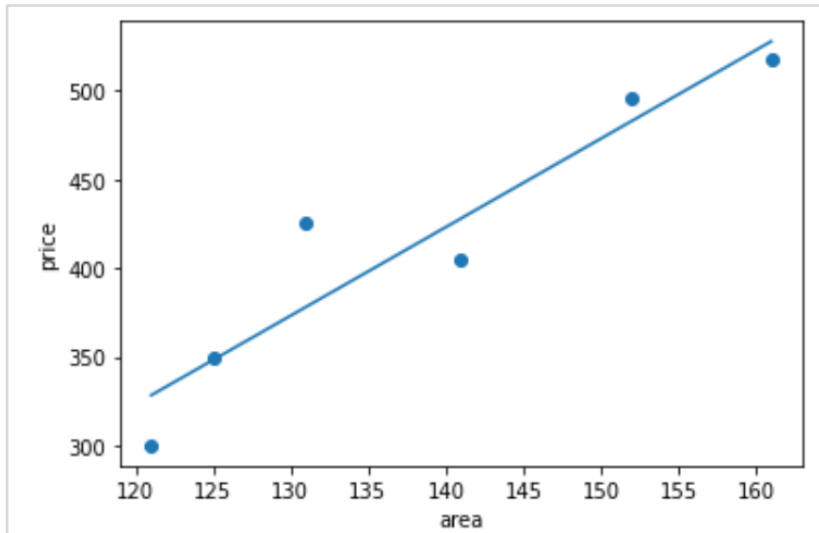
Output:

```
Slope: [4.98467124]
Intercept: -274.8769665187576
```

Input:

```
plt.scatter(x,y)
plt.xlabel("area") # X axis indicates the area.
plt.ylabel("price") # Y axis indicates the price.
plt.plot([x[0],x[-1]], [x[0]*w+b,x[-1]*w+b])
```

Output:



Step 5 Start a prediction task using the model.

Input:

```
testX = np.array([[130]]) # A test sample with an area of 130
lr.predict(testX)
```

Output:

```
Array([373.13029447]) # The model predicts the house price of the sample.
```

1.2.2 Linear Regression Implementation (Expansion Experiment)

This experiment uses the `lr_data.txt` dataset, which contains simulated house area and price data. To obtain the dataset, see the Lab Environment Setup.

Step 1 Import dependencies.

Input:

```
import numpy as np
import matplotlib.pyplot as plt
```

Step 2 Define the function for calculating gradients.

Input:

```
def generate_gradient(X, theta, y):
    sample_count = X.shape[0]
    # Calculate the gradient based on the matrix 1/m Σ(((h(x^i)-y^i)) x_j^i)
    return (1./sample_count)*X.T.dot(X.dot(theta)-y)
```

Step 3 Define the function for reading datasets.

Input:

```
def get_training_data(file_path):
    orig_data = np.loadtxt(file_path, skiprows=1) # Ignore the title in the first row of the dataset.
    cols = orig_data.shape[1]
```

```
return (orig_data, orig_data[:, :cols - 1], orig_data[:, cols-1:])
```

Step 4 Define the function for initializing parameters.

Input:

```
# Initialize the  $\theta$  array.
def init_theta(feature_count):
    return np.ones(feature_count).reshape(feature_count, 1)
```

Step 5 Define the function for implementing gradient descent.

Input:

```
def gradient_descending(X, y, theta, alpha):
    Jthetas= [] # Record the change trend of the cost function  $J(\theta)$  to confirm the gradient descent is correct.
    # Calculate the loss function, which is equal to the square of the difference between the actual value and the predicted value:  $(y^i - h(x^i))^2$ 
    Jtheta = (X.dot(theta)-y).T.dot(X.dot(theta)-y)
    index = 0
    gradient = generate_gradient(X, theta, y) # Calculate the gradient.
    while not np.all(np.absolute(gradient) <= 1e-5): # End the calculation when the gradient is less than 0.00001.
        theta = theta - alpha * gradient
        gradient = generate_gradient(X, theta, y) # Calculate the new gradient.
        # Calculate the loss function, which is equal to the square of the difference between the actual value and the predicted value:  $(y^i - h(x^i))^2$ 
        Jtheta = (X.dot(theta)-y).T.dot(X.dot(theta)-y)
        if (index+1) % 10 == 0:
            Jthetas.append((index, Jtheta[0])) # Record the result every 10 calculations.
        index += 1
    return theta, Jthetas
```

Step 6 Define the function for visualizing the change curve of the loss function.

Input:

```
# Plot the loss function change curve.
def showJTheta(diff_value):
    p_x = []
    p_y = []
    for (index, sum) in diff_value:
        p_x.append(index)
        p_y.append(sum)
    plt.plot(p_x, p_y, color='b')
    plt.xlabel('steps')
    plt.ylabel('loss function')
    plt.title('step - loss function curve')
    plt.show()
```

Step 7 Define the function for visualizing data points and the fitted curve.

Input:

```
# Plot the actual data points and the fitted curve.
def showlinecurve(theta, sample_training_set):
    x, y = sample_training_set[:, 1], sample_training_set[:, 2]
    z = theta[0] + theta[1] * x
    plt.scatter(x, y, color='b', marker='x', label="sample data")
    plt.plot(x, z, 'r', color="r", label="regression curve")
    plt.xlabel('x')
    plt.ylabel('y')
    plt.title('liner regression curve')
    plt.legend()
    plt.show()
```

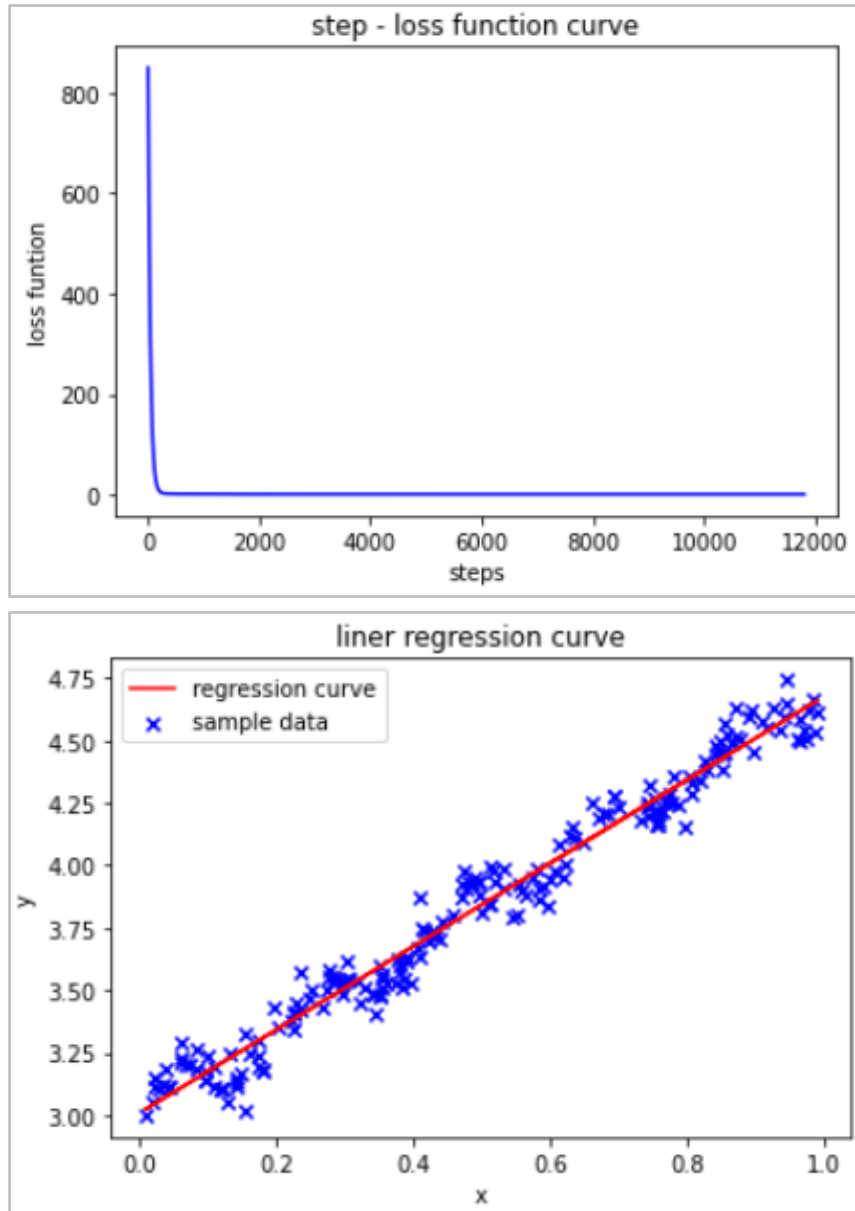
Step 8 Plot the final results.

Input:

```
# Read the dataset.
training_data_include_y, training_x, y = get_training_data("./ML/02/lr2_data.txt")
# Obtain the numbers of samples and features, respectively.
sample_count, feature_count = training_x.shape
# Define the learning step  $\alpha$ .
alpha = 0.01
# Initialize  $\theta$ .
theta = init_theta(feature_count)
# Obtain the final parameter  $\theta$  and cost.
result_theta, Jthetas = gradient_descending(training_x, y, theta, alpha)
# Display the parameter.
print("w:{} ".format(result_theta[0][0]), "b:{} ".format(result_theta[1][0]))
showJTheta(Jthetas)
showlinecurve(result_theta, training_data_include_y)
```

Output (ignore warnings):

```
w:3.0076279423997594 b:1.668677412281192
```



1.2.3 Logistic Regression

This experiment uses a custom dataset of house rent and area. The dataset is defined during the initial phase of the experiment.

Step 1 Import dependencies.

Input:

```
# Import StandardScaler from sklearn.preprocessing.
from sklearn.preprocessing import StandardScaler
# Import LogisticRegression from sklearn.linear_model.
from sklearn.linear_model import LogisticRegression
```

Step 2 Define the dataset.

Input:

```
# Each item in X denotes the rent and area.
# y indicates whether to rent the room (0: no; 1: yes).
X=[[2200,15],[2750,20],[5000,40],[4000,20],[3300,20],[2000,10],[2500,12],[12000,80],
   [2880,10],[2300,15],[1500,10],[3000,8],[2000,14],[2000,10],[2150,8],[3400,20],
   [5000,20],[4000,10],[3300,15],[2000,12],[2500,14],[10000,100],[3150,10],
   [2950,15],[1500,5],[3000,18],[8000,12],[2220,14],[6000,100],[3050,10]
  ]

y=[1,1,0,0,1,1,1,1,0,1,1,0,1,1,0,1,0,0,0,1,1,1,0,1,0,1,0,1,0,1,1,0]
```

Step 3 Preprocess data.

Standardize data to ensure that the variance of feature data in each dimension is 1 and the mean is 0. In this way, the prediction result is not dominated by large feature values of some dimensions.

Input:

```
ss = StandardScaler()
X_train = ss.fit_transform(X)
```

Display the standardized data.

Input:

```
print(X_train)
```

Output:

```
[[-0.60583897 -0.29313058]
 [-0.37682768 -0.09050576]
 [ 0.56003671  0.71999355]
 [ 0.14365254 -0.09050576]
 [-0.14781638 -0.09050576]
 [-0.68911581 -0.49575541]
 [-0.48092372 -0.41470548]
 [ 3.47472592  2.34099218]
 [-0.32269773 -0.49575541]
 [-0.56420055 -0.29313058]
 [-0.89730789 -0.49575541]
 [-0.27273163 -0.57680534]
 [-0.68911581 -0.33365555]
 [-0.68911581 -0.49575541]
 [-0.62665818 -0.57680534]
 [-0.10617796 -0.09050576]
 [ 0.56003671 -0.09050576]
 [ 0.14365254 -0.49575541]
 [-0.14781638 -0.29313058]
 [-0.68911581 -0.41470548]
 [-0.48092372 -0.33365555]
 [ 2.64195758  3.15149149]
 [-0.21027401 -0.49575541]
 [-0.29355084 -0.29313058]
```

```
[-0.89730789 -0.69838024]
[-0.27273163 -0.17155569]
[ 1.80918923 -0.41470548]
[-0.59751129 -0.33365555]
[ 0.97642089  3.15149149]
[-0.25191242 -0.49575541]]
```

Step 4 Fit the data.

Input:

```
# Use the fit method of LogisticRegression to train model parameters.
lr = LogisticRegression()
lr.fit(X_train, y)
```

Output:

```
LogisticRegression()
```

Step 5 Predict the data.

Input:

```
testX = [[2000,8]]
X_test = ss.transform(testX)
print("Value to be predicted: ",X_test)
label = lr.predict(X_test)
print("predicted label = ", label)
# Output the predicted probability.
prob = lr.predict_proba(X_test)
print("probability = ",prob)
```

Output:

```
Value to be predicted: [[-0.68911581 -0.57680534]]
predicted label = [1]
probability = [[0.41886952 0.58113048]]
```

1.2.4 Decision Tree

This experiment uses the **tennis.txt** dataset, which contains 14 samples. Each sample contains weather-related features and whether it is suitable for tennis.

Step 1 Import dependencies.

Input:

```
import pandas as pd
import numpy as np
from sklearn import tree
import pydotplus
```

Step 2 Define the function for generating a decision tree.

Input:

```
# Generate a decision tree.
def createTree(trainingData):
    data = trainingData.iloc[:, :-1] # Feature matrix
    labels = trainingData.iloc[:, -1] # Labels
    trainedTree = tree.DecisionTreeClassifier(criterion="entropy") # Decision tree classifier
    trainedTree.fit(data, labels) # Train the model.
    return trainedTree
```

Step 3 Define the function for saving the generated tree diagram.

Input:

```
def showtree2pdf(trainedTree, finename):
    dot_data = tree.export_graphviz(trainedTree, out_file=None) # Export the tree in Graphviz format.
    graph = pydotplus.graph_from_dot_data(dot_data)
    graph.write_pdf(finename) # Save the tree diagram to the local machine in PDF format.
```

Step 4 Define the function for generating vectorized data.

In the function, **pd.Categorical(list).codes** obtains the sequence number list corresponding to the original data, so as to convert the categorical information into numeric information.

Input:

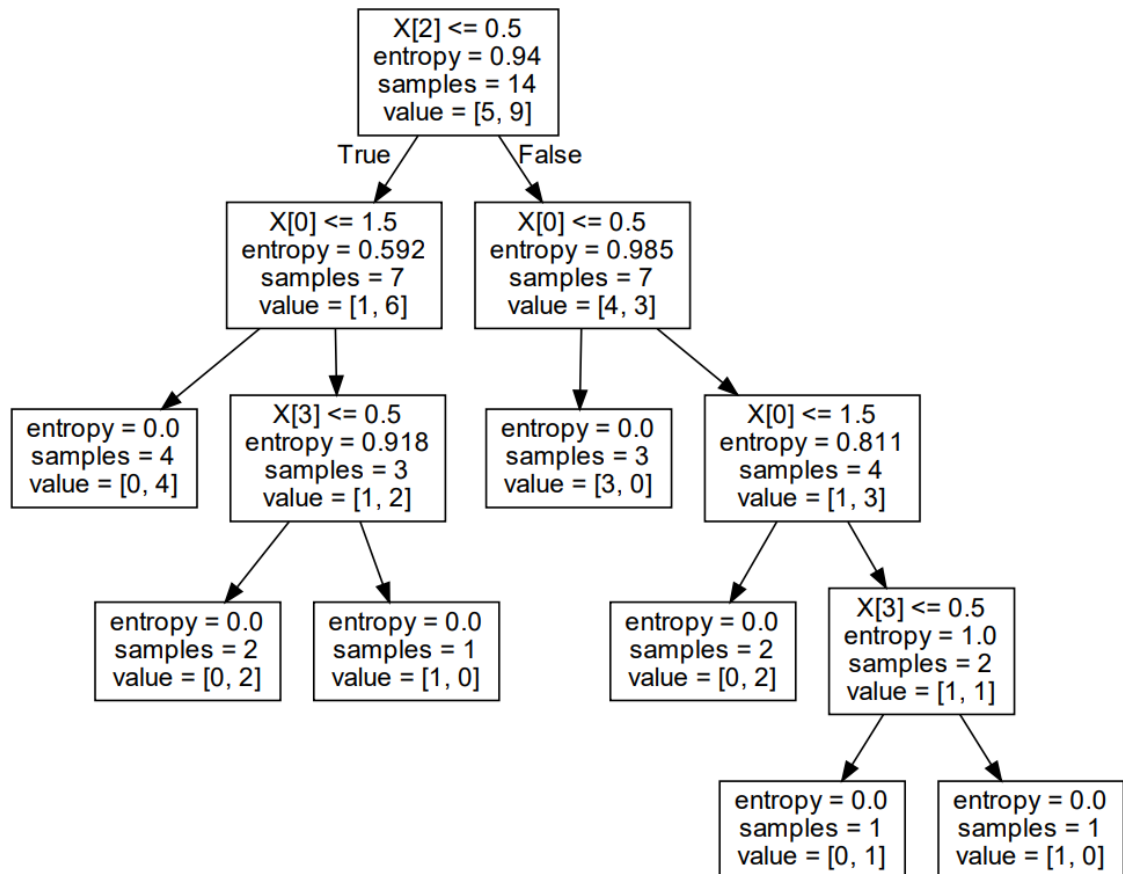
```
def data2vectoc(data):
    names = data.columns[:-1]
    for i in names:
        col = pd.Categorical(data[i])
        data[i] = col.codes
    return data
```

Step 5 Invoke the function for prediction.

Input:

```
data = pd.read_table("./ML/tennis.txt", header=None, sep='\t') # Read training data.
trainingvec = data2vectoc(data) # Vectorize data.
decisionTree = createTree(trainingvec) # Create a decision tree.
showtree2pdf(decisionTree, "tennis.pdf") # Plot the decision tree.
```

A decision tree diagram named **tennis.pdf** is generated on the local machine.



The file content is a visualized display of the decision tree. In the diagram, **X[2]** is the third feature variable (humidity); **X[0]** is the first feature variable (weather); **X[3]** is the fourth feature variable (wind); **entropy** is the entropy value of the node; and **samples** is the number of samples in the node, for example, 14 in the first node (root node) indicates the number of samples in the training set; and **value** indicates the numbers of samples of different types, for example, in the root node, 5 indicates the number of "no" samples, and 9 indicates the number of "yes" samples.

Predict a new sample. Input:

```
testVec = [0,0,1,1] # Weather is sunny, temperature is low, humidity is high, and wind is strong.
print(decisionTree.predict(np.array(testVec).reshape(1,-1))) # Predict.
```

Output:

```
['N']
```

1.2.5 K-means Algorithm Implementation

Step 1 Import dependencies.

make_blobs is used to generate the dataset required for this experiment, **matplotlib** is used for data visualization, and **KMeans** is used for fitting data based on the K-means algorithm.

Input:

```
from sklearn.datasets import make_blobs
```



```
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

Step 2 Generate the dataset.

make_blobs() is used to generate a dataset for a clustering task by returning the generated dataset and labels. Note: the K-means algorithm is often used to process unlabeled datasets in production. The generated dataset contains labels only to facilitate understanding.

Input:

```
X, y = make_blobs(n_samples=500,n_features=2,centers=4,random_state=1)
```

n_samples is the number of samples, **n_features** is the number of features of each sample (the dimension of data), **centers** is the number of categories, and **random_state** is the seed for the random number generator. The seed ensures that data generated each time is the same.

Display the dimension information of the generated dataset.

Input:

```
Print("Dimension of X is {}".format(X.shape))
Print("Dimension of y is {}".format(y.shape))
```

Output:

```
Dimension of X is (500, 2)
Dimension of y is (500,).
```

The dataset contains 500 sample points, and each sample point contains two features.

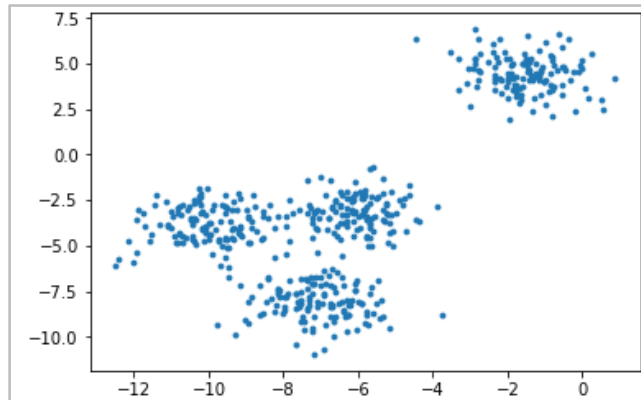
Step 3 Draw scatter graphs.

First, directly draw a scatter graph of the dataset. The labels generated during dataset generation are not considered for now. The graph contains four different clusters.

Input:

```
fig, ax1 = plt.subplots(1)
ax1.scatter(X[:, 0], X[:, 1]
            ,marker='o' # Set the shape of the point to circle.
            ,s=8 # Set the size of the point.
            )
plt.show()
```

Output:



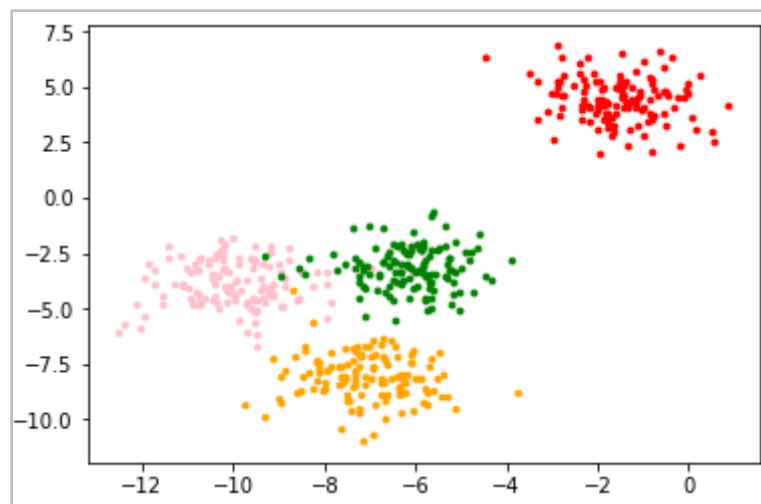
Next, draw a scatter graph where points in different colors correspond to the generated labels.

Input:

```
color = ["red","pink","orange","green"]
fig, ax1 = plt.subplots(1)

for i in range(4):
    ax1.scatter(X[y==i, 0], X[y==i, 1] # Draw the color based on the label.
               ,marker='o' # Set the shape of the point to circle.
               ,s=8 # Set the size of the point.
               ,c=color[i]
    )
plt.show()
```

Output:



Step 4 Perform k-means clustering.

Use **sklearn.cluster.KMeans** provided by scikit-learn to implement K-means clustering. First, aggregate the data samples into three types.

Input:

```
n_clusters = 3
cluster1 = KMeans(n_clusters=n_clusters,random_state=3).fit(X)
```

View the label of each sample point after clustering.

Input:

```
y_pred1 = cluster1.labels_
print(y_pred1)
```

Output:

```
[0 0 2 1 2 1 2 2 2 2 0 0 2 1 2 0 2 0 1 2 2 2 2 1 2 2 1 1 2 2 0 1 2 0 2 0 2
 2 0 2 2 2 1 2 2 0 2 2 1 1 1 2 2 2 0 2 2 2 2 2 1 1 2 2 1 2 0 2 2 2 0 2 2 0
 2 2 0 2 2 2 1 1 2 1 1 2 2 1 2 2 1 0 2 2 1 0 0 2 0 1 1 0 1 2 1 2 2 1 1 2 2
 0 1 2 1 2 1 2 1 2 2 0 0 2 2 2 1 0 0 2 1 2 2 2 2 0 1 2 1 1 2 0 2 1 1 1 2 2
 0 0 2 2 1 0 1 2 2 2 2 2 2 2 2 1 0 0 0 2 1 0 2 2 0 1 2 2 2 2 0 2 2 1 0 0
 2 2 0 0 2 1 1 0 0 2 1 2 0 0 1 0 2 1 2 2 0 2 2 0 2 2 2 2 0 2 2 2 1 2 1 2 0
 2 2 2 2 2 1 2 1 0 2 0 2 1 1 2 0 1 0 2 2 0 0 0 0 2 2 0 2 2 1 1 2 2 1 2 2 2
 1 2 1 2 2 1 2 0 0 2 2 2 2 1 1 2 1 2 0 1 0 1 0 0 1 0 1 1 2 2 2 2 2 2 0 1
 0 0 0 2 2 2 0 2 0 0 2 0 0 2 1 0 2 2 1 1 2 0 1 1 2 0 1 1 2 2 1 2 2 0 0 1 2
 0 2 1 1 2 2 2 0 2 1 1 2 1 1 1 1 0 0 2 1 2 2 0 1 2 1 2 1 2 2 2 1 2 2 0 1 0
 0 0 0 0 0 2 0 1 0 1 1 2 1 2 2 2 0 1 2 1 2 0 2 2 0 2 2 1 1 0 2 2 1 2 2 0 0
 2 0 2 2 0 2 0 2 1 0 1 2 2 1 2 2 1 0 2 1 1 2 2 2 2 0 1 0 2 1 0 0 0 2 1 2 0
 2 2 2 2 0 2 2 2 2 2 0 2 2 0 2 1 2 1 2 2 2 1 1 1 2 2 2 0 2 1 2 0 1 0 1 0
 2 1 1 0 2 2 0 2 2 2 0 2 1 2 2 0 0 0 2]
```

Print the centroid of each cluster.

Input:

```
centroid1 = cluster1.cluster_centers_
print(centroid1)
```

Output:

```
[[-7.09306648 -8.10994454]
 [-1.54234022  4.43517599]
 [-8.0862351  -3.5179868 ]]
```

Visualize the clustering result.

Input:

```
color = ["red", "pink", "orange", "gray"]

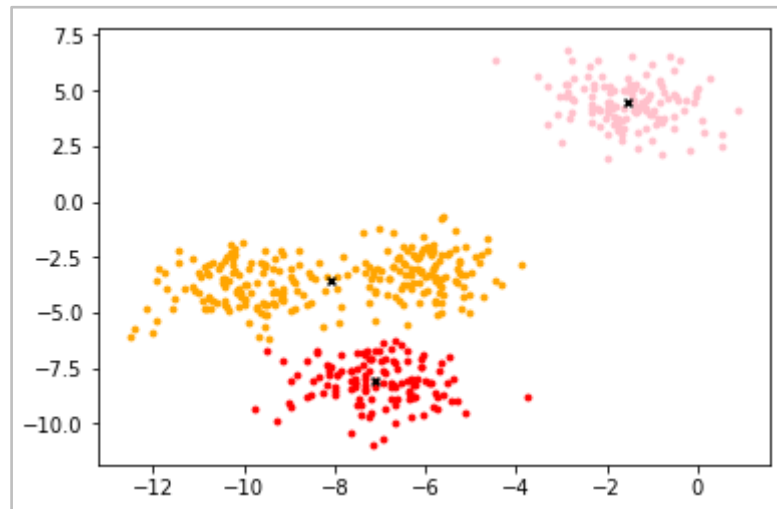
fig, ax1 = plt.subplots(1)

for i in range(n_clusters):
    ax1.scatter(X[y_pred1==i, 0], X[y_pred1==i, 1],
               ,marker='o' # Set the shape of the point to circle.
               ,s=8 # Set the size of the point.
               ,c=color[i]
               )

ax1.scatter(centroid1[:,0],centroid1[:,1],
           ,marker="x"
           ,s=15
           ,c="black")
```

```
plt.show()
```

Output:



The graph shows that the data samples are aggregated into three clusters, and the centroid of each cluster is represented by x.

Step 5 Perform k-means clustering again.

Next, perform the preceding steps to use k-means clustering to aggregate data samples into four types.

Input:

```
n_clusters = 4
cluster2 = KMeans(n_clusters=n_clusters,random_state=0).fit(X)
y_pred2 = cluster2.labels_
centroid2 = cluster2.cluster_centers_
print("Centroid: {}".format(centroid2))
```

Output:

```
Centroid: [[ -6.08459039  -3.17305983]
 [ -1.54234022   4.43517599]
 [ -7.09306648  -8.10994454]
 [-10.00969056  -3.84944007]]
```

Visualize the clustering result.

Input:

```
color = ["red","pink","orange","green"]

fig, ax1 = plt.subplots(1)

for i in range(n_clusters):
    ax1.scatter(X[y_pred2==i, 0], X[y_pred2==i, 1]
                ,marker='o' # Set the shape of the point to circle.
                ,s=8 # Set the size of the point.
                ,c=color[i])
```

```

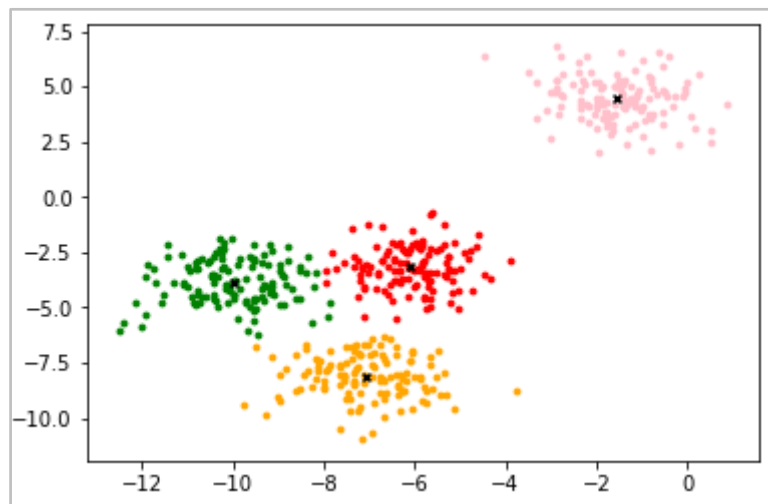
)

ax1.scatter(centroid2[:,0],centroid2[:,1]
            ,marker="x"
            ,s=15
            ,c="black")

plt.show()

```

Output:



The graph shows that the data samples are aggregated into four clusters, and the centroid of each cluster is represented by x. Compare the scatter graphs generated based on the original data and data aggregated into four types. It can be seen that many sample points are aggregated into wrong clusters.

1.3 Question

How to implement K-means from scratch using Python?

1.4 Summary

This lab introduces the machine learning implementation process, including data import, segmentation, standardization, as well as models and hyperparameters. Common machine learning algorithms are implemented based on scikit-learn to help you better understand how to build and use machine learning models.

Huawei Certification-AI Series Training

HCIA-AI

Deep Learning and AI Development Framework Lab Guide

Issue: 4.0



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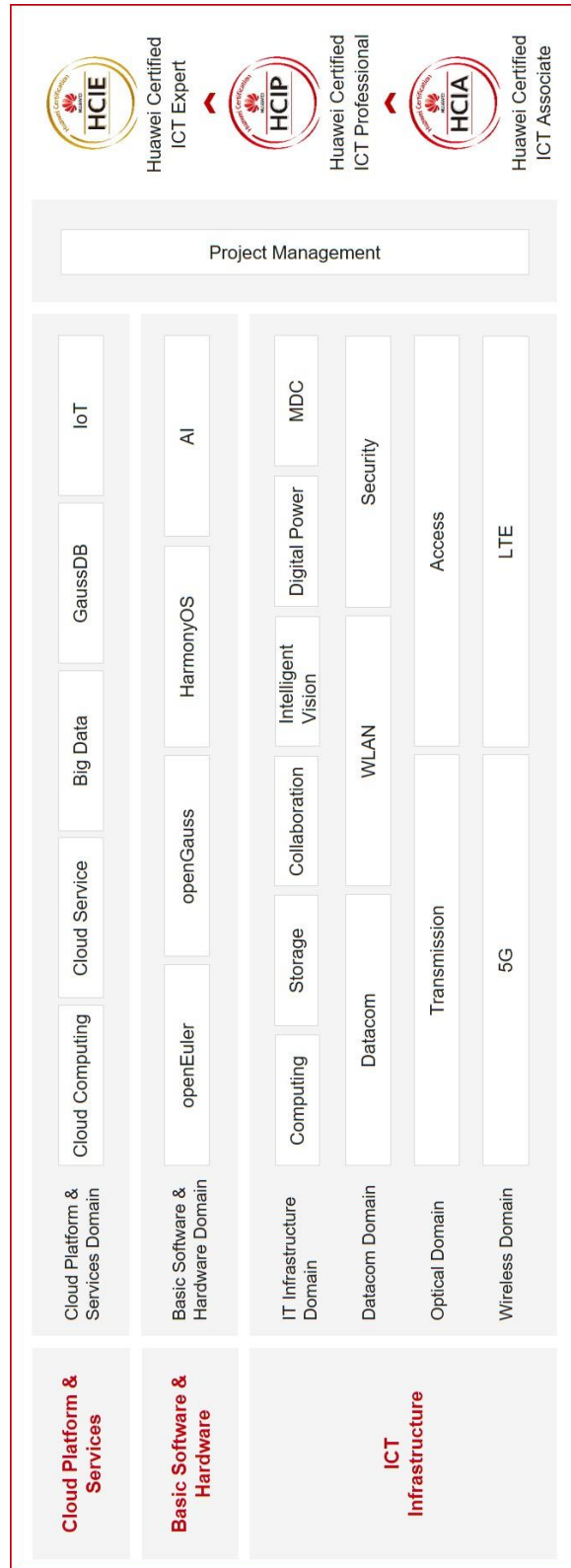
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About This Document

Overview

This document is applicable to the candidates who are preparing for the HCIA-AI exam or readers who want to understand AI basics and basics of PyTorch programming.

Description

This lab guide introduces the following one experiment:

- Use Python and basic third-party libraries to implement a three-layer fully-connected neural network, including the network structure, gradient calculation, and parameter optimization. Use different optimizers for training and compare the loss function changes. This experiment provides reference for trainees to understand the overall process of deep learning and the basic implementation of each phase.

Link to the data and code files used in the experiment:

<https://certification-data.obs.cn-north-4.myhuaweicloud.com/ENG/HCIA-AI/V4.0/Dataset.zip>

Background Knowledge Required

This document is for Huawei's basic certification. To better understand this document, ensure that you meet the following requirements:

- Have basic knowledge about Python.
-

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1 Building a Fully-Connected Neural Network from Scratch

1.1 Experiment Introduction

1.1.1 About This Experiment

This experiment aims to use the Python programming language and basic libraries such as NumPy to build a multi-layer fully-connected neural network, and use the MNIST dataset for training and testing to recognize handwritten digital images.

The purpose of the experiment is to build, train, and test neural networks without using any deep learning framework. By gradually implementing each function, you can gain a deep understanding of the implementation principle and training and testing process of neural networks.

The MNIST dataset is a widely used handwritten digit recognition dataset. It contains a large number of handwritten digit images which are used to train and test various image processing and recognition technologies, especially handwritten digit recognition.

The MNIST dataset consists of handwritten images of 10 digits from 0 to 9, and is divided into a training set and a test set. The training set contains 60,000 samples, and the test set contains 10,000 samples. Each sample is a 28 x 28 pixel grayscale image with pixel values ranging from 0 (white) to 255 (black). These images were extracted from a project of the National Institute of Standards and Technology (NIST). The purpose of this project is to collect a large amount of handwritten digital data for the research and development of optical character recognition (OCR) systems.

1.1.2 File Parsing

Decompress the file after downloading the code and dataset files. The file directories are as follows:

```
|— MNIST
|   |— functions.py
|   |— gradient.py
|   |— layers.py
|   |— load_mnist.py
|   |— multi_layer_net.py
|   |— train_mnist.py
|   |— optimizer.py
```

```
| |— util.py
| |— t10k-images-idx3-ubyte
| |— t10k-labels-idx1-ubyte
| |— train-images-idx3-ubyte
| |— train-labels-idx1-ubyte
```

Data files:

train-images-idx3-ubyte: image data of the training set, including 60,000 28 x 28 pixel grayscale images.

train-labels-idx1-ubyte: image labels of the training set, including 60,000 digital labels.

t10k-images-idx3-ubyte: image data of the test set, including 10,000 28 x 28 pixel grayscale images.

t10k-labels-idx1-ubyte: image labels of the test set, including 10,000 digital labels.

mnist.pkl: serialized by the Python pickle module and contains the training set and validation set of the MNIST dataset.

train_mnist.py: code entry for the experiment, and the main training program.

load_mnist.py: contains implementation of the data loading function.

multi_layer_net.py: implementation of a multi-layer neural network.

layers.py: implementation of basic layer functions.

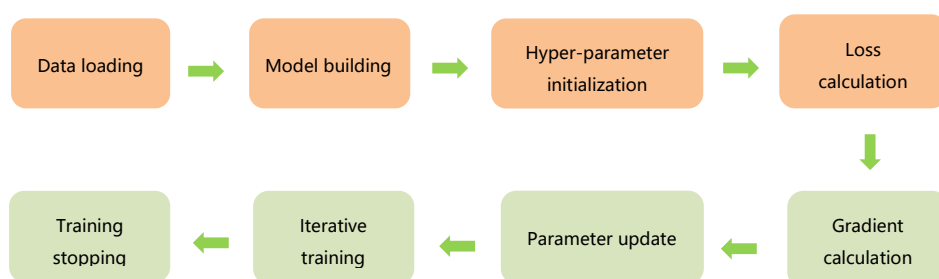
gradient.py: calculates the implementation of the gradient function.

optimizer.py: contains implementation of multiple optimizers.

util.py: contains implementation of the data conversion function.

functions.py: contains implementation of multiple activation functions.

1.1.3 Experiment Process



1.2 Training Code Parsing

1.2.1 Main Program

train_mnist.py is the main program of the project. The code is as follows:

```
# coding: utf-8
import matplotlib.pyplot as plt
from load_mnist import load_mnist
from util import smooth_curve
from multi_layer_net import MultiLayerNet
from optimizer import *

# 0: Read the MNIST data=====
(x_train, t_train), (x_test, t_test) = load_mnist(normalize=True)

train_size = x_train.shape[0]
batch_size = 128
max_iterations = 1000

# 1: Set the experiment=====
optimizers = {} # Create a dictionary for storing optimizers.
optimizers['SGD'] = SGD()
optimizers['Momentum'] = Momentum()
optimizers['AdaGrad'] = AdaGrad()
optimizers['Adam'] = Adam()

networks = {} # Create a dictionary for storing network structures.
train_loss = {} # Create a dictionary for storing loss functions.
for key in optimizers.keys():
    networks[key] = MultiLayerNet(
        input_size=784, hidden_size_list=[100, 100, 100, 100],
        output_size=10)
    train_loss[key] = []

# 2: Start training=====
for i in range(max_iterations):
    batch_mask = np.random.choice(train_size, batch_size)
    x_batch = x_train[batch_mask]
    t_batch = t_train[batch_mask]

    for key in optimizers.keys():
        grads = networks[key].gradient(x_batch, t_batch)
        optimizers[key].update(networks[key].params, grads)

        loss = networks[key].loss(x_batch, t_batch)
        train_loss[key].append(loss)

    if i % 100 == 0:
        print( "===== " + "iteration:" + str(i) + "=====")
```

```

        for key in optimizers.keys():
            loss = networks[key].loss(x_batch, t_batch)
            print(key + ":" + str(loss))

# 3: Draw a graph=====
markers = {"SGD": "o", "Momentum": "x", "AdaGrad": "s", "Adam": "D"}
x = np.arange(max_iterations)
for key in optimizers.keys():
    plt.plot(x, smooth_curve(train_loss[key]), marker=markers[key], markevery=100, label=key)
plt.xlabel("iterations")
plt.ylabel("loss")
plt.ylim(0, 1)
plt.legend()
plt.show()

```

1.2.2 Loading the Dataset

The compressed package of the MNIST dataset has been downloaded. Use the `load_mnist` function to load the dataset. This function loads the dataset to the pickle file. To read the dataset, you only need to load the pickle file.

```

# Read the MNIST data.
(x_train, t_train), (x_test, t_test) = load_mnist(normalize=True)

train_size = x_train.shape[0] # Obtain the number of training images.
batch_size = 128             # Set batch_size.
max_iterations = 1000        # Set the maximum number of iterations.

```

Analyze the `load_mnist` function.

Open the `load_mnist.py` file.

The `load_mnist.py` file declares the following variables:

```

# key_file stores the name of the dataset package as a dictionary.
key_file = {
    'train_img': 'train-images-idx3-ubyte.gz',
    'train_label': 'train-labels-idx1-ubyte.gz',
    'test_img': 't10k-images-idx3-ubyte.gz',
    'test_label': 't10k-labels-idx1-ubyte.gz'
}

# The dataset_dir variable is used to obtain the absolute path of the folder where the current code file is
located.
dataset_dir = os.path.dirname(os.path.abspath(__file__))

# save_file declares the path of the .pkl file through character string concatenation.
save_file = dataset_dir + "/mnist.pkl"

```

The body of the `load_mnist` function is as follows:

```

def load_mnist(normalize=True, flatten=True, one_hot_label=False):
    """Read the MNIST dataset.
    Parameters

```

```

-----
normalize: normalizes the pixel value of the image to a value within the range from 0.0 to 1.0.
one_hot_label :
    When one_hot_label is set to True, the label is returned as a one-hot array.
    A one-hot array is similar to [0,0,1,0,0,0,0,0,0].
flatten: indicates whether to expand the image into a one-dimensional array.

Returns
-----
(training image, training label), (test image, test label)
"""
# Check whether the pickle file for storing the dataset exists. If the file does not exist, use the compressed
dataset to create the file.
if not os.path.exists(save_file):
    init_mnist()
# Use the pickle file to load the dataset.
with open(save_file, 'rb') as f:
    dataset = pickle.load(f)

"""Preprocess the dataset"""
if normalize:
    for key in ('train_img', 'test_img'):
        dataset[key] = dataset[key].astype(np.float32) # Convert to the np.float32 data type.
        dataset[key] /= 255.0 # Convert the pixel value to a value within the range of 0 to 1.

if one_hot_label: # Convert the label to a one-hot array.
    dataset['train_label'] = _change_one_hot_label(dataset['train_label'])
    dataset['test_label'] = _change_one_hot_label(dataset['test_label'])

if not flatten: # Check whether the data needs to be flattened. If the data needs to be flattened, the
(784,) format is retained. Otherwise, the data is converted into (28, 28).
    for key in ('train_img', 'test_img'):
        dataset[key] = dataset[key].reshape(-1, 1, 28, 28)

return (dataset['train_img'], dataset['train_label']), (dataset['test_img'], dataset['test_label'])

```

Analyze other functions called in the load_mnist function body.

init_mnist function:

```

def init_mnist():
    dataset = _convert_numpy () # Convert the dataset into the NumPy format.
    print("Creating pickle file ...")
    with open(save_file, 'wb') as f: # Open the .pkl file in binary writeable mode.
        pickle.dump (dataset, f, -1) # Write the dataset into the .pkl file. After the operation is complete,
the file is automatically closed.
    print("Done!")

```

_convert_numpy function:

```

def _convert_numpy():
    dataset = {} # Create an empty dictionary.
    # Save the dataset in the dictionary by name.
    dataset['train_img'] = _load_img(key_file['train_img'])
    dataset['train_label'] = _load_label(key_file['train_label'])

```



```
dataset['test_img'] = _load_img(key_file['test_img'])
dataset['test_label'] = _load_label(key_file['test_label'])

return dataset
```

`_load_label` and `_load_img` functions:

```
# Load the label.
def _load_label(file_name):
    file_path = dataset_dir + "/" + file_name # Absolute path of the file to which the label is written.

    print("Converting " + file_name + " to NumPy Array ...")
    with gzip.open(file_path, 'rb') as f: # Open the compressed dataset.
        """
        np.frombuffer():
        This is a function in the NumPy library and is used to create a NumPy array from the byte buffer.
        The first parameter is the buffer. In this example, it reads the file content through f.read ().
        f.read():
        f is an opened file object.
        f.read () is a file object method. It reads all the remaining content of a file and returns the content
        in bytes.
        np.uint8:
        Specifies that the data in the buffer read by the np.frombuffer () function should be interpreted
        as uint8.
        offset=8:
        This parameter specifies the number of bytes at the beginning of the buffer that should be
        skipped before data parsing.
        It specifies that the function should skip the first eight bytes of the buffer and start creating an
        array from the ninth byte.
        """
        labels = np.frombuffer(f.read(), np.uint8, offset=8)
        print("Done")

    return labels
# The structure of the _load_img function is the same as that of the _load_label function. The _load_img
function is used to load data (images).
def _load_img(file_name):
    file_path = dataset_dir + "/" + file_name

    print("Converting " + file_name + " to NumPy Array ...")
    with gzip.open(file_path, 'rb') as f:
        data = np.frombuffer(f.read(), np.uint8, offset=16)
        data = data.reshape(-1, 784)
        print("Done")

    return data
```

`_change_one_hot_label` function:

```
def _change_one_hot_label(X):
    T = np.zeros((X.size, 10)) # Initialize the all-zero matrix T with the shape of (X.size, 10).
    for idx, row in enumerate(T):
        row[X[idx]] = 1 # Set the corresponding label position in one-hot to 1.
    # T = np.eye(10)[X] #The preceding code can be replaced with this row.
```

```
return T
```

1.2.3 Building the Network Structure

```
# 1. Set multiple optimizers and analyze the loss changes when different optimizers are used.
optimizers = {}
optimizers['SGD'] = SGD()
optimizers['Momentum'] = Momentum()
optimizers['AdaGrad'] = AdaGrad()
optimizers['Adam'] = Adam()

# Build the network model.
networks = {} # Save the network model.
train_loss = {} # Create a dictionary to store the loss values of different optimizers.
for key in optimizers.keys():
    networks[key] = MultiLayerNet(
        input_size=784, hidden_size_list=[100, 100, 100, 100],
        output_size=10)
    train_loss[key] = []
```

Parse the MultiLayerNet class.

```
class MultiLayerNet:
    """Fully-connected multi-layer neural network

    Parameters
    -----
    input_size: input size (784 for MNIST)
    hidden_size_list: list of the number of neurons at the hidden layer (for example, [100, 100, 100])
    output_size: output size (10 for MNIST)
    activation : 'relu' or 'sigmoid'
    weight_init_std: standard deviation of the weight (for example, 0.01).
        Set the initial value of he when relu or he is specified.
        Set the initial value of xavier when sigmoid or xavier is specified.
    weight_decay_lambda: strength of weight decay (L2 norm)
    """
    def __init__(self, input_size, hidden_size_list, output_size,
                  activation='relu', weight_init_std='relu', weight_decay_lambda=0):
        #Configure the network.
        self.input_size = input_size
        self.output_size = output_size
        self.hidden_size_list = hidden_size_list
        self.hidden_layer_num = len(hidden_size_list)
        self.weight_decay_lambda = weight_decay_lambda
        self.params = {}

        # Initialize the weight.
        self.__init_weight(weight_init_std)

        # Generate the layers.
        activation_layer = {'sigmoid': Sigmoid, 'relu': Relu}
        self.layers = OrderedDict()
        for idx in range(1, self.hidden_layer_num+1):
            self.layers['Affine' + str(idx)] = Affine(self.params['W' + str(idx)],
```

```

        self.params['b' + str(idx)]
        self.layers['Activation_function' + str(idx)] = activation_layer[activation]()

    idx = self.hidden_layer_num + 1
    self.layers['Affine' + str(idx)] = Affine(self.params['W' + str(idx)],
        self.params['b' + str(idx)])

    self.last_layer = SoftmaxWithLoss()

def __init_weight(self, weight_init_std):
    """Set the initial weight.

    Parameters
    -----
    weight_init_std: standard deviation of the weight (for example, 0.01).
        Set the initial value of he when relu or he is specified.
        Set the initial value of xavier when sigmoid or xavier is specified.
    """
    all_size_list = [self.input_size] + self.hidden_size_list + [self.output_size]
    for idx in range(1, len(all_size_list)):
        scale = weight_init_std
        if str(weight_init_std).lower() in ('relu', 'he'):
            scale = np.sqrt(2.0 / all_size_list[idx - 1]) # Recommended initial value when ReLU is
used.
        elif str(weight_init_std).lower() in ('sigmoid', 'xavier'):
            scale = np.sqrt(1.0 / all_size_list[idx - 1]) # Recommended initial value when sigmoid is
used.

        self.params['W' + str(idx)] = scale * np.random.randn(all_size_list[idx-1], all_size_list[idx])
        self.params['b' + str(idx)] = np.zeros(all_size_list[idx])

def predict(self, x):
    for layer in self.layers.values():
        x = layer.forward(x)

    return x

def loss(self, x, t):
    """Function for calculating the loss
    Parameters
    -----
    x: input data
    t: correct label

    Returns
    -----
    Value of the loss function
    """
    y = self.predict(x)

    weight_decay = 0
    for idx in range(1, self.hidden_layer_num + 2):
        W = self.params['W' + str(idx)]
        weight_decay += 0.5 * self.weight_decay_lambda * np.sum(W ** 2)

```

```

        return self.last_layer.forward(y, t) + weight_decay

def accuracy(self, x, t):
    y = self.predict(x)
    y = np.argmax(y, axis=1)
    if t.ndim != 1 : t = np.argmax(t, axis=1)

    accuracy = np.sum(y == t) / float(x.shape[0])
    return accuracy

def numerical_gradient(self, x, t):
    """Gradient calculation (numerical differentiation)
    Parameters
    -----
    x: input data
    t: correct label

    Returns
    -----
    Dictionary variable with the gradient of each layer.
        grads['W1'], grads['W2']...: weight of each layer
        grads['b1'], grads['b2']...: offset of each layer
    """
    loss_W = lambda W: self.loss(x, t)
    grads = {}
    for idx in range(1, self.hidden_layer_num+2):
        grads['W' + str(idx)] = numerical_gradient(loss_W, self.params['W' + str(idx)])
        grads['b' + str(idx)] = numerical_gradient(loss_W, self.params['b' + str(idx)])

    return grads

def gradient(self, x, t):
    """Gradient calculation (error backpropagation method)
    Parameters
    -----
    x: input data
    t: teacher label

    Returns
    -----
    Dictionary variable with the gradient of each layer.
        grads['W1'], grads['W2']...: weight of each layer
        grads['b1'], grads['b2']...: offset of each layer
    """
    # forward
    self.loss(x, t)

    # backward
    dout = 1
    dout = self.last_layer.backward(dout)

    layers = list(self.layers.values())
    layers.reverse()
    for layer in layers:
        dout = layer.backward(dout)

```

```
# Settings
grads = {}
for idx in range(1, self.hidden_layer_num+2):
    grads['W' + str(idx)] = self.layers['Affine' + str(idx)].dW + self.weight_decay_lambda *
self.layers['Affine' + str(idx)].W
    grads['b' + str(idx)] = self.layers['Affine' + str(idx)].db

return grads
```

1.2.4 Starting Training

```
# 2: Start training=====
for i in range(max_iterations):
    batch_mask = np.random.choice(train_size, batch_size) # Generate a random number.
    x_batch = x_train[batch_mask] # Use a random number as the subscript and select the training
data.
    t_batch = t_train[batch_mask] # Use a random number as the subscript and select the
corresponding label.

    # Parameter optimization
    for key in optimizers.keys():
        grads = networks[key].gradient(x_batch, t_batch)
        optimizers[key].update(networks[key].params, grads)

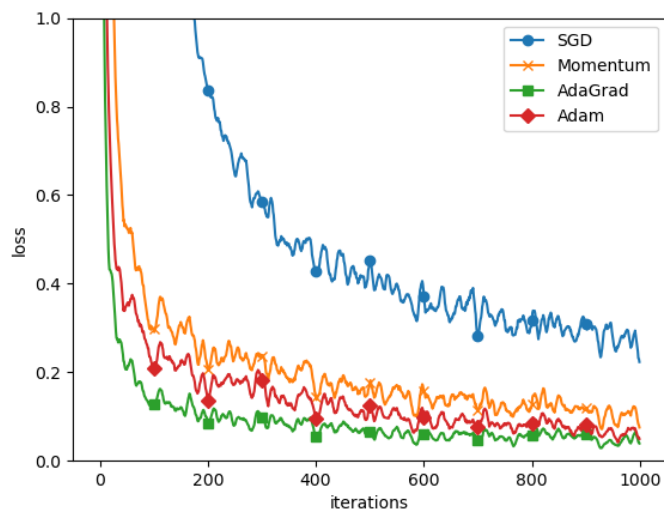
        loss = networks[key].loss(x_batch, t_batch)
        train_loss[key].append(loss)

    if i % 100 == 0:
        print( "======" + "iteration:" + str(i) + "======" )
        for key in optimizers.keys():
            loss = networks[key].loss(x_batch, t_batch)
            print(key + ":" + str(loss))
```

1.2.5 Drawing the Graph

```
markers = {"SGD": "o", "Momentum": "x", "AdaGrad": "s", "Adam": "D"}
x = np.arange(max_iterations)
for key in optimizers.keys():
    plt.plot(x, smooth_curve(train_loss[key]), marker=markers[key], markevery=100, label=key)
plt.xlabel("iterations")
plt.ylabel("loss")
plt.ylim(0, 1)
plt.legend()
plt.show()
```

1.2.6 Experiment Result



1.3 Questions

1. How do I change the learning rate in the code? Use different learning rate setting policies and observe the loss curve.
 2. Add a test module to the code to generate the test accuracy in different phases.
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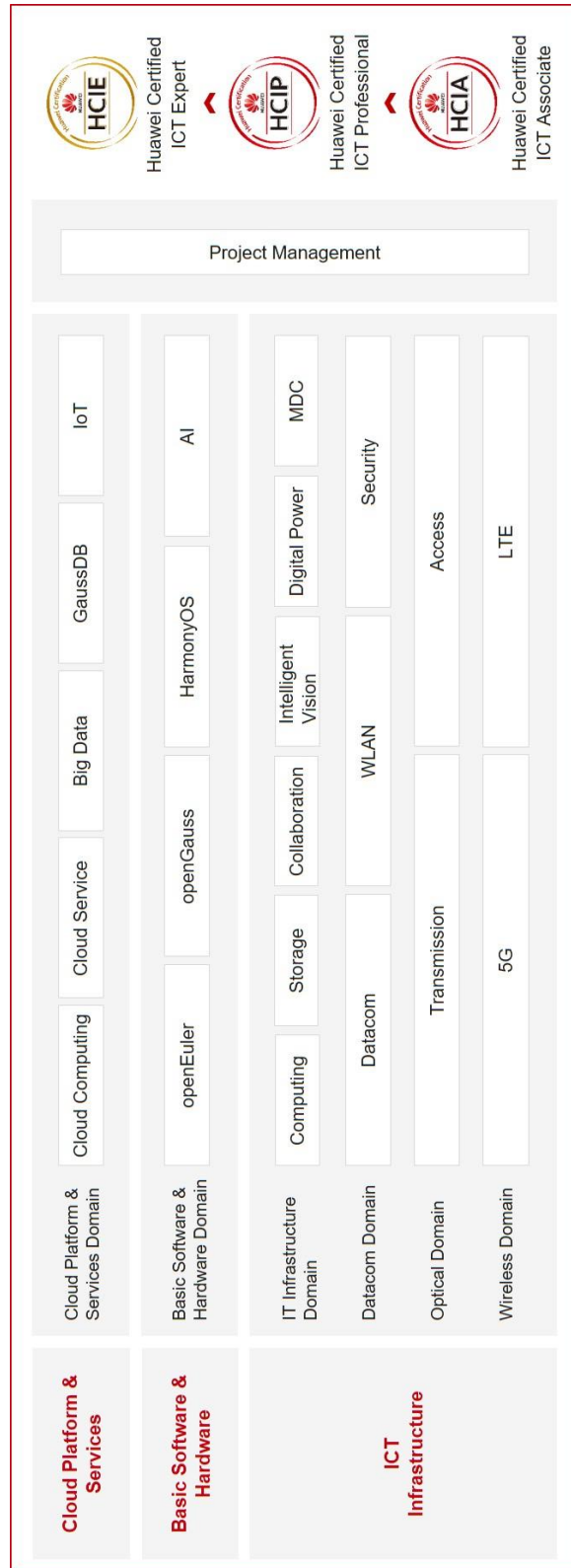
Understand the history of AI development.

Master algorithms related to traditional machine learning and deep learning.

Be capable of building, training, and deploying neural networks by using development framework PyTorch.

Be competent in sales, marketing, product manager, project management, technical support, and other AI positions.

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About This Document

Overview

This document is applicable to the candidates who are preparing for the HCIA-AI exam or readers who want to understand AI basics and basics of PyTorch programming.

Description

This lab guide introduces the following one experiments:

- Experiment 1 describes how to use PyTorch. At the end of the experiment, you will be able to use PyTorch for image classification and text classification.

Link to the data and code files used in the experiment:

<https://certification-data.obs.cn-north-4.myhuaweicloud.com/ENG/HCIA-AI/V4.0/Dataset.zip>

Background Knowledge Required

This document is for Huawei's basic certification. To better understand this document, ensure that you meet the following requirements:

- Have basic knowledge about Python.
-

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1 PyTorch

1.1 Experiment Introduction

1.1.1 About This Experiment

This experiment introduces the tensor data structure of PyTorch. By performing operations on tensors, you can understand the basic PyTorch syntax. After learning about basic operations, you can use PyTorch to build the LeNet and ResNet network structures to implement image classification.

1.1.2 Experiment Objectives

- To understand how to create a PyTorch tensor
- To understand the attributes and methods of PyTorch tensors
- To learn to build PyTorch networks

1.1.3 Experiment Environment

The PyTorch framework is required for this experiment. The recommended version is 2.0 or later. You can conduct the experiment on your PC or log in to Huawei Cloud and purchase ModelArts.

1.2 PyTorch Basics

1.2.1 Creating a Tensor

Tensors of different dimensions represent different data. A 0-dimensional tensor represents a scalar, a 1-dimensional tensor represents a vector, a 2-dimensional tensor represents a matrix, and a 3-dimensional tensor may represent three RGB channels of a color image, and the like.

Step 1 Randomly initialize a tensor.

Code:

```
import torch
x = torch.rand(4, 3)
print(x)
```

Output:

```
tensor([[0.2052, 0.2865, 0.9629],
        [0.3342, 0.8409, 0.0419],
        [0.8176, 0.9898, 0.8704],
        [0.0259, 0.0501, 0.3116]])
```

Step 2 Create a tensor with different types of data.

Code:

```
import numpy as np
print(torch.Tensor(np.array([[1, 2], [3, 4]]))) # Use an array to create a tensor.
print(torch.Tensor([True,False])) # Use bool data to create a tensor.
print(torch.Tensor((1,2,3,4,5))) # Use a tuple to create a tensor.
print(torch.Tensor([i for i in range(10)])) # Use a list to create a tensor.
```

Output:

```
tensor([[1., 2.],
        [3., 4.]])
tensor([1., 0.])
tensor([1., 2., 3., 4., 5.])
tensor([0., 1., 2., 3., 4., 5., 6., 7., 8., 9.])
```

Step 3 Build different types of tensors.

Code:

```
print (torch.FloatTensor (2,3).type ()) # Build a 2 x 3 tensor of the float type.
print (torch.DoubleTensor (2,3).type ()) # Build a 2 x 3 tensor of the double type.
print (torch.HalfTensor (2,3).type ()) # Build a 2 x 3 tensor of the HalfTensor type.
print (torch.ByteTensor (2,3).type ()) # Build a 2 x 3 tensor of the byte type.
print (torch.CharTensor (2,3).type ()) # Build a 2 x 3 tensor of the char type.
print (torch.ShortTensor (2,3).type ()) # Build a 2 x 3 tensor of the short type.
print (torch.IntTensor (2,3).type ()) # Build a 2 x 3 tensor of the Int type.
print (torch.LongTensor (2,3).type ()) # Build a 2 x 3 tensor of the long type.
```

Output:

```
torch.FloatTensor
torch.DoubleTensor
torch.HalfTensor
torch.ByteTensor
torch.CharTensor
torch.ShortTensor
torch.IntTensor
torch.LongTensor
```

Step 4 Create special tensors.

Code:

```
import torch
x = torch.zeros(4, 3, dtype=torch.long)
```

```
print(x) # All-zero tensor
x = torch.ones(4, 3, dtype=torch.long)
print(x) # All-1 tensor
x = x.new_ones(4, 3, dtype=torch.double)
print(x)
x = torch.randn_like(x, dtype=torch.float)
print(x) #
```

Output:

```
tensor([[0, 0, 0],
        [0, 0, 0],
        [0, 0, 0],
        [0, 0, 0]])
tensor([[1, 1, 1],
        [1, 1, 1],
        [1, 1, 1],
        [1, 1, 1]])
tensor([[1., 1., 1.],
        [1., 1., 1.],
        [1., 1., 1.],
        [1., 1., 1.]], dtype=torch.float64)
tensor([[ -0.8838,  0.8029, -1.9473],
        [-0.7147, -1.0430, -0.3002],
        [ 1.2838,  1.9056, -1.6879],
        [-0.8628,  1.6469,  0.6585]])
```

1.2.2 Tensor Attributes

Tensor attributes include shape and data type (dtype).

- The tensor shape is a tuple.
- The tensor dtype is a data type of PyTorch.

Code:

```
x = torch.Tensor(np.array([[1, 2], [3, 4]]))
print(x.shape) # Obtain the shape.
print(x.size())
print(x.dim()) # Number of dimensions
print(x.device) # Hardware
print(x.dtype) # Data type
```

Output:

```
torch.Size([2, 2])
torch.Size([2, 2])
2
cpu
torch.float32
```

1.2.3 Tensor Method

Code:

```
x = torch.Tensor(np.array([[1, 2], [3, 4]]))

print (x.reshape (4,1)) # reshape
x = x.unsqueeze (2) # Add a dimension.
print(x.shape)
x = x.squeeze (2) # Compress dimensions.
print(x.shape)

print (x.flatten ()) # Flatten to one dimension.
x.to (device) # Move to the specified device.
```

Output:

```
tensor([[1.,
         2.,
         3.,
         4.]])
torch.Size([2, 2, 1])
torch.Size([2, 2])
tensor([1., 2., 3., 4.]])
```

1.2.4 Loading the Dataset

Step 1 Load the MNIST dataset.

Before loading the dataset, download the MNIST dataset from <http://yann.lecun.com/exdb/mnist/> and store the training and test files in the MNIST folder. If you do not download the dataset, PyTorch automatically downloads it. Ensure that the network connection is normal.

Code:

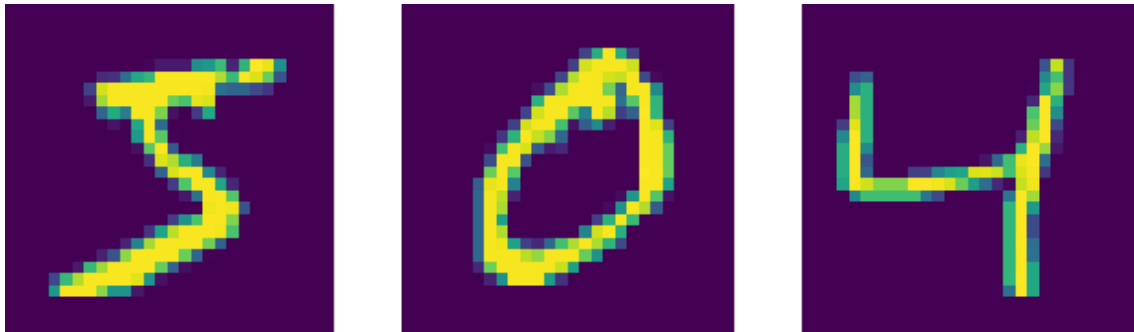
```
import matplotlib.pyplot as plt
from torchvision import datasets, transforms

# Read images from the MNIST dataset.
train_dataset =
datasets.MNIST(root='./MNIST',train=True,download=True,transform=transforms.ToTensor())
# View the image and set the image size.
plt.figure(figsize=(8,8))
i = 1

# Print three subplots of the training set.
for dic in train_dataset.train_data[:3]:
    plt.subplot(3,3,i)
    plt.imshow(dic)
    plt.axis('off')
    i +=1
plt.show()
```

Output:

Figure 1-1 Samples of the MNIST dataset



Step 2 Use ImageFolder to load the dataset.

The flower dataset used in the ResNet experiment is loaded in this experiment.

```
train = torchvision.datasets.ImageFolder("./flower_photos_train/") # Load the data.

# Check the dataset attributes.
print("Dataset types")
print(train.classes)
print(train.class_to_idx)
print("Dataset samples")
print(train.samples[0])
print("Image data format")
print(train.extensions[0])
```

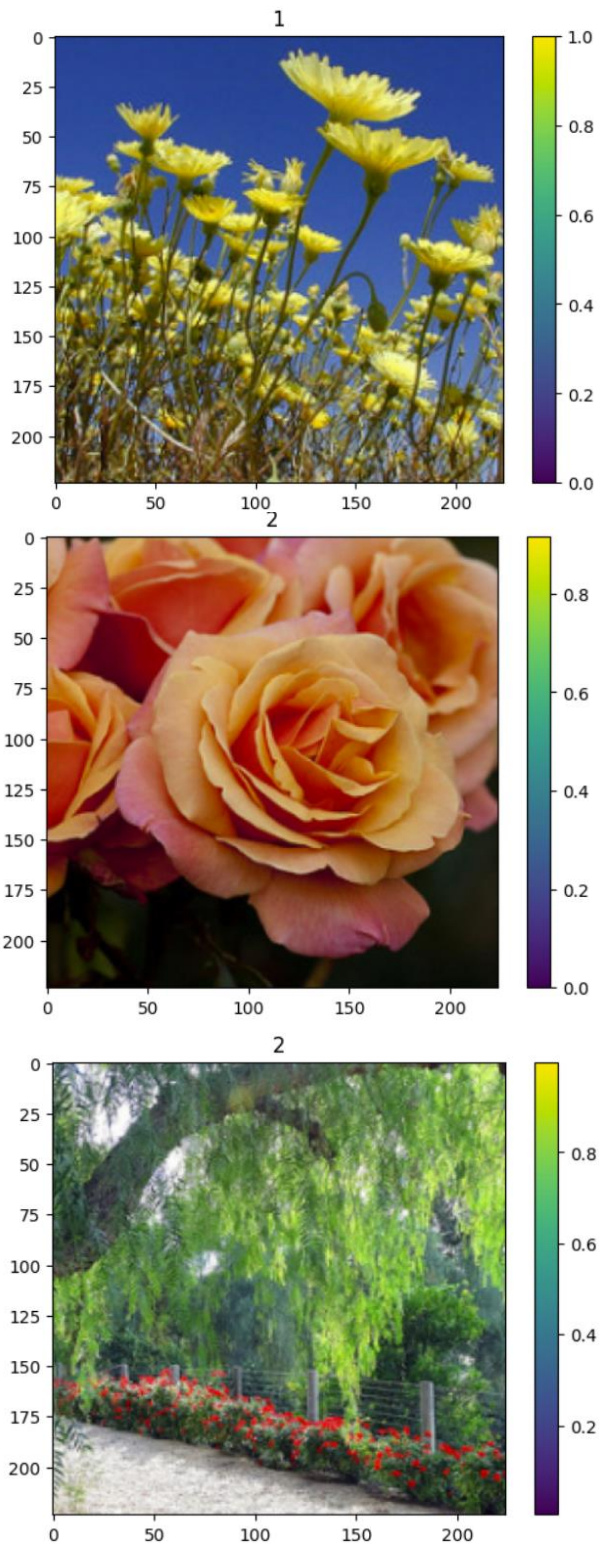
Output:

```
Dataset types
['daisy', 'dandelion', 'roses', 'sunflowers', 'tulips']
{'daisy': 0, 'dandelion': 1, 'roses': 2, 'sunflowers': 3, 'tulips': 4}
Dataset samples
('./flower_photos_train/daisy\\10140303196_b88d3d6cec.jpg', 0)
Image data format
.jpg
```

Visualize the data.

```
# Display the loaded image data.
for i in range(5):
    plt.figure()
    random.seed = 2
    num = random.randint(0,3000)
    plt.imshow(train[num][0].permute(1,2,0)) # train[num] returns a tuple containing data and the
category number corresponding to the label.
    plt.title(train[num][1])
    plt.colorbar()
    plt.grid(False)
plt.show()
```

Output:



Step 3 Customize the dataset.

In PyTorch, a custom dataset class usually needs to inherit the `torch.utils.data.Dataset` class and implement the `__len__` and `__getitem__` methods. The two methods are used to return

the number of samples in the dataset and load and preprocess the samples of a given index.

```
import os
import torch
from torch.utils.data import Dataset, DataLoader
from PIL import Image
from torchvision import transforms
# Pseudocode
class CustomDataset(Dataset):
    def __init__(self, image_dir, label_file, transform=None):
        """
        image_dir (string): directory for storing images.
        label_file (string): path of the label file. Each row corresponds to a label.
        transform (callable, optional): optional image conversion operation, which is used for data
        augmentation or preprocessing. The default value is None.
        """
        self.image_dir = image_dir # Image directory
        self.labels = self._load_labels(label_file) # Load labels.
        self.image_filenames = sorted(os.listdir(image_dir)) # Obtain the image file name list and sorts
        the files.
        self.transform = transform # Image transformation

    def _load_labels(self, label_file):
        """
        Load labels from a file.
        Args:
            label_file (string): path of the label file
        Returns:
            list: Label list. Each element is an integer.
        """
        with open(label_file, 'r') as f:
            labels = [int(line.strip()) for line in f] # Read the file, convert each row into an integer, and
            store the integers in the list.
        return labels

    def __len__(self):
        """
        Return the number of samples in the dataset.
        Returns: number of samples
        """
        return len(self.image_filenames) # Return the length of the image file name list.

    def __getitem__(self, idx):
        """
        Obtain samples in the dataset based on indexes.
        Args:
            idx (int): sample index
        Returns:
            tuple: a tuple containing images and labels
        """
        img_name = os.path.join(self.image_dir, self.image_filenames[idx]) # Build the full path of the
        image file.
        image = Image.open(img_name).convert('RGB') # Open the image file and convert it to the
        RGB mode.
```

```
label = self.labels[idx] # Obtain the corresponding label.

if self.transform:
    image = self.transform(image) # Transform the image if the transformation operation is
allowed.

# Return the tuple of the images and labels.
return image, label

if __name__ == "__main__":
    # Define the image transformation operation.
    transform = transforms.Compose([
        # Data processing
        pass
    ])

    # Image and label file paths
    image_dir = 'path/to/images' # Replace it with your image file path.
    label_file = 'path/to/labels.txt' # Replace it with your label file path.

    # Create the dataset instance.
    dataset = CustomDataset(image_dir=image_dir, label_file=label_file, transform=transform)

    # Create the data loader.
    dataloader = DataLoader(dataset, batch_size=32, shuffle=True, num_workers=4) # Set the batch
size to 32, shuffle the data sequence, and use four working processes.
```

1.3 LeNet

Use PyTorch to implement the LeNet network.

Step 1 Import the toolkit.

```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
import os
import matplotlib.pyplot as plt
```

Step 2 Define the data mode and use it to load data.

The loaded dataset is CIFAR-10, which contains RGB color images of 10 categories: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. Each category has 6000 images. There is a total of 60,000 images in the dataset, 50,000 of which are used for training and 10,000 used for testing. The size of each image is 32 x 32 pixels.

```
# Data preprocessing
transform = transforms.Compose(
    [transforms.ToTensor(),
     transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5))])
```

```
# Download and load the CIFAR-10 dataset.
trainset = torchvision.datasets.CIFAR10(root='./data', train=True, download=True, transform=transform)
trainloader = torch.utils.data.DataLoader(trainset, batch_size=64, shuffle=True, num_workers=1)

testset = torchvision.datasets.CIFAR10(root='./data', train=False, download=True, transform=transform)
testloader = torch.utils.data.DataLoader(testset, batch_size=64, shuffle=False, num_workers=1)
```

Step 3 Display the data.

```
# Display the images.
dataiter = iter(trainloader)
images, labels = next(dataiter)
# Create a grid to display images.
imshow = torchvision.utils.make_grid(images)

# Convert to a NumPy array and remove normalization.
imshow = imshow.numpy().transpose((1, 2, 0))

# Show images.
plt.imshow(imshow)
plt.show()
```

Step 4 Define and instantiate a network structure class.

```
# Define the LeNet network structure.
class LeNet(nn.Module):
    def __init__(self):
        super(LeNet, self).__init__() # Call the constructor of the parent class nn.Module.
        # For the first convolutional layer, the number of input channels is 3, the number of output
        # channels is 6, and the size of the convolution kernel is 5 x 5.
        self.conv1 = nn.Conv2d(3, 6, kernel_size=5)
        # For the second convolutional layer, the number of input channels is 6, the number of output
        # channels is 16, and the size of the convolution kernel is 5 x 5.
        self.conv2 = nn.Conv2d(6, 16, kernel_size=5)
        # For the first fully-connected layer, the number of input features is 16 x 5 x 5, and the number of
        # output features is 120.
        self.fc1 = nn.Linear(16*5*5, 120)
        # For the second fully-connected layer, the number of input features is 120, and the number of
        # output features is 84.
        self.fc2 = nn.Linear(120, 84)
        # For the third fully-connected layer (output layer), the number of input features is 84, and the
        # number of output features is 10.
        self.fc3 = nn.Linear(84, 10)

    def forward(self, x):
        x = torch.tanh(self.conv1(x))
        # Perform 2 x 2 max pooling on x.
        x = torch.max_pool2d(x, 2)
        # x passes through the second convolutional layer, and then applies the tanh activation function.
        x = torch.tanh(self.conv2(x))
        # Perform 2 x 2 max pooling on x.
        x = torch.max_pool2d(x, 2)
```

```
# Flatten x into a one-dimensional vector and input it to the fully-connected layer. The size is
(batch_size, 16 x 5 x 5).
x = x.view(-1, 16*5*5)
x = torch.tanh(self.fc1(x))
# x passes through the second fully-connected layer and then applies the tanh activation
function.
x = torch.tanh(self.fc2(x))
# x passes through the third fully-connected layer (output layer). Generally, the activation
function is not used here because the output value may be used for tasks such as classification.
x = self.fc3(x)
return x

# Instantiate the LeNet network.
net = LeNet()
net.parameters # View the model parameters.
```

Output:

```
<bound method Module.parameters of LeNet(
  (conv1): Conv2d(3, 6, kernel_size=(5, 5), stride=(1, 1))
  (conv2): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
  (fc1): Linear(in_features=400, out_features=120, bias=True)
  (fc2): Linear(in_features=120, out_features=84, bias=True)
  (fc3): Linear(in_features=84, out_features=10, bias=True)
)>
```

Step 5 Define the loss function and optimizer.

```
criterion = nn.CrossEntropyLoss()
optimizer = optim.SGD(net.parameters(), lr=0.001, momentum=0.9)
```

Step 6 Define the training process.

```
# Check whether there are trained parameters to load.
if os.path.isfile("cifar10.pth"):
    net.load_state_dict(torch.load("cifar10.pth"))
# Store the loss value.
loss_history = []

# Train the network.
for epoch in range(50): # 50 epochs
    running_loss = 0.0
    for i, data in enumerate(trainloader, 0):
        inputs, labels = data

        optimizer.zero_grad()

        outputs = net(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item()
    if i % 200 == 199:
```

```
print(f'Epoch: {epoch + 1}, Batch: {i + 1} loss: {running_loss / 200:.3f}')
running_loss = 0.0

# Record the loss value.
loss_history.append(loss.item())

print('Finished Training')
```

Step 7 Visualize the loss decrease process.

```
# Draw the loss curve.
plt.plot(loss_history)
plt.xlabel('Batch')
plt.ylabel('Loss')
plt.title('Loss over Training Batches')
plt.show()
```

Step 8 Use the test set to test the model.

```
# Test the performance of the network with the test set.
correct = 0
total = 0
with torch.no_grad():
    for data in testloader:
        images, labels = data
        outputs = net(images)
        _, predicted = torch.max(outputs.data, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

print(f'Accuracy of the network on the 10000 test images: {100 * correct / total:.2f}%')
```

Output:

```
Accuracy of the network on the 10000 test images: 75.97%
```

1.4 ResNet-50 Image Classification

The purpose of this experiment is ResNet-50 image classification, a typical case of deep learning. The process is as follows:

- Process the required dataset. The flower image dataset is used here.
- Define a network. Here we need to build a ResNet-50 model.
- Define the loss function and optimizer.
- Load the dataset and perform training. After the training is complete, use the test set for evaluation.

Step 1 Prepare for the experiment.

Before you start, check whether PyTorch is correctly installed.

In addition, you shall have basic mathematical knowledge such as Python coding basics, probability, and matrix.

Recommended environment:

Version: PyTorch 2.5

Programming language: Python 3 (≥ 3.7)

Step 2 Prepare data.

A flower image dataset is used in this example. It is an open-source dataset that includes five types of flowers: daisy (633 images), dandelion (898 images), roses (641 images), sunflowers (699 images), and tulips (799 images). The 3670 images with a size of about 230 MB are stored in five directories. To test the model after it is deployed, the dataset is divided into two parts: `flower_photos_train` and `flower_photos_test`.

The directory structure is as follows (in the `flowers` folder of the dataset):

`flower_photos_train`

└─ daisy

└─ dandelion

└─ roses

└─ sunflowers

└─ tulips

└─ LICENSE.txt

`flower_photos_test`

└─ daisy

└─ dandelion

└─ roses

└─ sunflowers

└─ tulips

└─ LICENSE.txt

Download the data file.

```
!pip install download
```

```
from download import download
url_train =
b"aHR0cHM6Ly9hc2NlbmQtcHJvZmVzc2lvbmFsLWNvbnN0cnVjdGlvbi1kYXRhc2V0Lm9icy5teWh1YXdlW
Nsb3VkLmNvbS9kZWVwLWxlYXJuaW5nL2Zsb3dlcl9waG90b3NfdHJhaW4uemlw"
url_test =
b"aHR0cHM6Ly9hc2NlbmQtcHJvZmVzc2lvbmFsLWNvbnN0cnVjdGlvbi1kYXRhc2V0Lm9icy5teWh1YXdlW
Nsb3VkLmNvbS9kZWVwLWxlYXJuaW5nL2Zsb3dlcl9waG90b3NfdGVzdC56aXA="
```

```
path_train = download(base64.b64decode(url_train).decode(), "./", kind="zip", replace=True,
verbose=False)
path_train = download(base64.b64decode(url_test).decode(), "./", kind="zip", replace=True,
verbose=False)
```

Step 3 Import the Python libraries and modules and configure running information.

Import the Python libraries before start.

```
import torch
import torchvision
import torchvision.transforms as transforms
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim
import matplotlib.pyplot as plt
from easydict import EasyDict as edict
import random
```

Step 4 Define the constant dictionary.

```
cfg = edict({
    'data_path': './flower_photos_train', # Path of the training dataset
    'test_path': 'flowers/flower_photos_train', # Path of the test dataset
    'data_size': 3616,
    'HEIGHT': 224, # Image height
    'WIDTH': 224, # Image width
    '_R_MEAN': 123.68, # Mean of CIFAR10
    '_G_MEAN': 116.78,
    '_B_MEAN': 103.94,
    '_R_STD': 1, # User-defined standard deviation
    '_G_STD': 1,
    '_B_STD': 1,
    '_RESIZE_SIDE_MIN': 256, # Minimum resize value of image enhancement
    '_RESIZE_SIDE_MAX': 512,

    'batch_size': 1, # Batch size
    'num_class': 5, # Classification
    'epoch_size': 5, # Number of training times
    'num_workers': 1,
    'device': "cpu", # cpu, xpu, etc.

    'prefix': 'renet50.pth', # Model name
})
```

Step 5 Read and process the data.

Define the data processing method.

```
# Use transforms.Compose to combine multiple processing modes.
transform_train = transforms.Compose([
```

```
transforms.RandomHorizontalFlip(),

transforms.ToTensor(),
transforms.Normalize(mean=[cfg._R_MEAN, cfg._G_MEAN, cfg._B_MEAN], std=[cfg._R_STD,
cfg._G_STD, cfg._B_STD]),
transforms.Resize(cfg._RESIZE_SIDE_MIN),
transforms.CenterCrop((cfg.HEIGHT, cfg.WIDTH)),

])

transform_test = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5)),
])
```

Load the data.

```
train = torchvision.datasets.ImageFolder(cfg. data_path, transform=transform_train)
trainloader = torch.utils.data.DataLoader(train, batch_size=cfg.batch_size, shuffle=True,
num_workers=cfg.num_workers)
```

Step 6 Define the network structure.

```
import torch.nn as nn

# Define the basic block, which is mainly used for ResNet-18 and ResNet-34.
class BasicBlock(nn.Module):
    expansion = 1 # Multiple for expansion of the number of output channels

    def __init__(self, in_channels, out_channels, stride=1):
        super(BasicBlock, self).__init__()

        # Residual function
        self.residual_function = nn.Sequential(
            nn.Conv2d(in_channels, out_channels, kernel_size=3, stride=stride, padding=1, bias=False),
            # The first convolutional layer
            nn.BatchNorm2d(out_channels), # The first batch normalization layer
            nn.ReLU(inplace=True), # Activation function
            nn.Conv2d(out_channels, out_channels * BasicBlock.expansion, kernel_size=3, padding=1,
bias=False), # The second convolutional layer
            nn.BatchNorm2d(out_channels * BasicBlock.expansion) # The second batch normalization
layer
        )

        # Shortcut, which is used to adjust the dimensions and stride
        self.shortcut = nn.Sequential()

        # If the dimensions do not match, use 1x1 convolution to adjust the dimensions of the shortcut.
        if stride != 1 or in_channels != BasicBlock.expansion * out_channels:
            self.shortcut = nn.Sequential(
                nn.Conv2d(in_channels, out_channels * BasicBlock.expansion, kernel_size=1,
stride=stride, bias=False),
                nn.BatchNorm2d(out_channels * BasicBlock.expansion)
            )
```

```

def forward(self, x):
    # Residual connection and activation function
    return nn.ReLU(inplace=True)(self.residual_function(x) + self.shortcut(x))

# Define the bottleneck block, which is mainly used for ResNet-50, ResNet-101, and ResNet-152.
class BottleNeck(nn.Module):
    expansion = 4 # Multiple for expansion of the number of output channels

    def __init__(self, in_channels, out_channels, stride=1):
        super(BottleNeck, self).__init__()

        # Residual function
        self.residual_function = nn.Sequential(
            nn.Conv2d(in_channels, out_channels, kernel_size=1, bias=False), # The first 1x1
convolutional layer for reducing dimensions
            nn.BatchNorm2d(out_channels), # The first batch normalization layer
            nn.ReLU(inplace=True), # Activation function
            nn.Conv2d(out_channels, out_channels, stride=stride, kernel_size=3, padding=1, bias=False),
# 3x3 convolutional layer
            nn.BatchNorm2d(out_channels), # The second batch normalization layer
            nn.ReLU(inplace=True), # Activation function
            nn.Conv2d(out_channels, out_channels * BottleNeck.expansion, kernel_size=1, bias=False),
# The second 1x1 convolutional layer for adding dimensions
            nn.BatchNorm2d(out_channels * BottleNeck.expansion) # The third batch normalization
layer
        )

        # Shortcut, which is used to adjust the dimensions and stride
        self.shortcut = nn.Sequential()

        # If the dimensions do not match, use 1x1 convolution to adjust the dimensions of the shortcut.
        if stride != 1 or in_channels != out_channels * BottleNeck.expansion:
            self.shortcut = nn.Sequential(
                nn.Conv2d(in_channels, out_channels * BottleNeck.expansion, stride=stride,
kernel_size=1, bias=False),
                nn.BatchNorm2d(out_channels * BottleNeck.expansion)
            )

    def forward(self, x):
        # Residual connection and activation function
        return nn.ReLU(inplace=True)(self.residual_function(x) + self.shortcut(x))

# Define the ResNet network.
class ResNet(nn.Module):

    def __init__(self, block, num_block, num_classes=100):
        super(ResNet, self).__init__()

        self.in_channels = 64 # Initial number of channels

        # The first convolutional layer
        self.conv1 = nn.Sequential(
            nn.Conv2d(3, 64, kernel_size=3, padding=1, bias=False),
            nn.BatchNorm2d(64),

```

```

        nn.ReLU(inplace=True)
    )

    # The number of layers varies depending on the ResNet version.
    self.conv2_x = self._make_layer(block, 64, num_block[0], 1)
    self.conv3_x = self._make_layer(block, 128, num_block[1], 2)
    self.conv4_x = self._make_layer(block, 256, num_block[2], 2)
    self.conv5_x = self._make_layer(block, 512, num_block[3], 2)

    # Global average pooling layer
    self.avg_pool = nn.AdaptiveAvgPool2d((1, 1))

    # Fully-connected layer. Set the number of output classes as required.
    self.fc = nn.Linear(512 * block.expansion, num_classes)

    def _make_layer(self, block, out_channels, num_blocks, stride):
        # Generate a sequence of layers.
        strides = [stride] + [1] * (num_blocks - 1) # Set the stride of each block to 1 except the first
        block.
        layers = []
        for stride in strides:
            layers.append(block(self.in_channels, out_channels, stride)) # Add a block.
            self.in_channels = out_channels * block.expansion # Update the number of channels.

        return nn.Sequential(*layers) # Return the layer sequence.

    def forward(self, x):
        # Forward propagation
        output = self.conv1(x)
        output = self.conv2_x(output)
        output = self.conv3_x(output)
        output = self.conv4_x(output)
        output = self.conv5_x(output)
        output = self.avg_pool(output) # Global average pooling
        output = output.view(output.size(0), -1) # Flatten into a one-dimensional vector.
        output = self.fc (output) # Fully-connected layer

        return output

```

Define the ResNet structure under different parameters (number of layers) based on the defined class.

```

# Define ResNets of different versions.
def resnet18():
    """Return the ResNet-18 model"""
    return ResNet(BasicBlock, [2, 2, 2, 2])

def resnet34():
    """Return the ResNet-34 model"""
    return ResNet(BasicBlock, [3, 4, 6, 3])

def resnet50():
    """Return the ResNet-50 model"""
    return ResNet(BottleNeck, [3, 4, 6, 3], num_classes=cfg.num_class) # Change the number of classes.

```

```
def resnet101():  
    """Return the ResNet-101 model"""  
    return ResNet(BottleNeck, [3, 4, 23, 3])  
  
def resnet152():  
    """Return the ResNet-152 model"""  
    return ResNet(BottleNeck, [3, 8, 36, 3])
```

ResNet-50 is used in the subsequent experiment.

Step 7 Define the loss function and optimizer, and initialize the network.

```
import os  
  
net = resnet50().to(cfg.device)  
if os.path.isfile(cfg.prefix):  
    net.load_state_dict(torch.load(cfg.prefix))  
  
criterion = nn.CrossEntropyLoss()  
optimizer = optim.SGD(net.parameters(), lr=0.1, momentum=0.9, weight_decay=0.0001)  
scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer, factor = 0.1, patience=5)
```

Step 8 Define the training process.

```
for epoch in range(cfg.epoch_size):  
    losses = []  
    running_loss = 0  
    for i, inp in enumerate(trainloader):  
        inputs, labels = inp  
        inputs, labels = inputs.to(cfg.device), labels.to(cfg.device)  
        optimizer.zero_grad()  
  
        outputs = net(inputs)  
        loss = criterion(outputs, labels)  
        losses.append(loss.item())  
  
        loss.backward()  
        optimizer.step()  
  
        running_loss += loss.item()  
  
        if i%200 == 0 and i > 0:  
            print(f'Loss [{epoch+1}, {i}](epoch, minibatch): ', running_loss / 100)  
            running_loss = 0.0  
  
    avg_loss = sum(losses)/len(losses)  
    scheduler.step(avg_loss)  
  
print('Training Done')
```

Step 9 Save the model.

```
torch.save(net.state_dict(),cfg.prefix)
```

Step 10 Test the model.

```
test = torchvision.datasets.ImageFolder(cfg.test_path, transform=transform_test)
testloader = torch.utils.data.DataLoader(test, batch_size=cfg.batch_size, shuffle=True,
num_workers=cfg.num_workers)

correct = 0
total = 0

with torch.no_grad():
    for data in testloader:
        images, labels = data
        images, labels = images.to(cfg.device), labels.to(cfg.device)
        outputs = net(images)

        _, predicted = torch.max(outputs.data, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()
print('Accuracy: ', 100*(correct/total), '%')
```

1.5 TextCNN Text Classification

Step 1 Create a notebook file and select the PyTorch environment.

Step 2 Import related libraries.

```
import os
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, Dataset
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
import numpy as np
import re
from collections import Counter
import urllib
import tarfile

# Set a random seed.
torch.manual_seed(42)
```

Step 3 Download the dataset.

```
# Download data.

def download_imdb(data_dir):
    # Create a data directory.
    if not os.path.exists(data_dir):
```

```
os.makedirs(data_dir)

# Download the IMDb dataset.
url = "https://ai.stanford.edu/~amaas/data/sentiment/acllmbd_v1.tar.gz"
file_path = os.path.join(data_dir, "acllmbd_v1.tar.gz")

if not os.path.exists(file_path):
    print("Downloading IMDB dataset...")
    urllib.request.urlretrieve(url, file_path)
    print("Download complete.")

# Decompress the file.
if not os.path.exists(os.path.join(data_dir, "acllmbd")):
    print("Extracting IMDB dataset...")
    with tarfile.open(file_path, "r:gz") as tar:
        tar.extractall(path=data_dir)
    print("Extraction complete.")

# Download and decompress the IMDb dataset.
data_dir = "data/imdb"
download_imdb(data_dir)
```

Step 4 Load data.

```
def load_imdb_data(data_dir):
    reviews = []
    labels = []

    for label_type in ['pos', 'neg']:
        dir_name = os.path.join(data_dir, 'acllmbd', 'train', label_type)
        for fname in os.listdir(dir_name):
            if fname.endswith('.txt'):
                with open(os.path.join(dir_name, fname), 'r', encoding='utf-8') as f:
                    reviews.append(f.read())
                    labels.append(1 if label_type == 'pos' else 0)

    return reviews, labels
```

Step 5 Preprocess text and build a vocabulary.

```
# Preprocess text.
def preprocess_text(text):
    text = re.sub(r'[\^\w\s]', "", text) # Removal of punctuations
    text = text.lower() # Conversion to lowercase
    return text

# Build a vocabulary.
def build_vocab(reviews, max_vocab_size=10000):
    word_counts = Counter()
    for review in reviews:
        word_counts.update(review.split())
    vocab = {word: i + 1 for i, (word, _) in enumerate(word_counts.most_common(max_vocab_size))}
    vocab['<PAD>'] = 0 # Addition of padding characters
    return vocab
```


Step 6 Convert the text into index sequences.

Input:

```
# Convert the text into index sequences.
def text_to_sequence(text, vocab):
    return [vocab.get(word, 0) for word in text.split()]

# Pad the sequences.
def pad_sequence(seq, max_len):
    if len(seq) >= max_len:
        return seq[:max_len]
    else:
        return seq + [0] * (max_len - len(seq))
```

Step 7 Define the dataset.

```
class IMDBDataset(Dataset):
    def __init__(self, X, y):
        self.X = X
        self.y = y

    def __len__(self):
        return len(self.X)

    def __getitem__(self, idx):
        return self.X[idx], self.y[idx]
```

Step 8 Define the TextCNN model.

```
class TextCNN(nn.Module):
    def __init__(self, vocab_size, embed_dim, num_classes, kernel_sizes=[3, 4, 5], num_filters=100):
        super(TextCNN, self).__init__()
        self.embedding = nn.Embedding(vocab_size, embed_dim)
        self.convs = nn.ModuleList([
            nn.Conv2d(1, num_filters, (k, embed_dim)) for k in kernel_sizes
        ])
        self.fc = nn.Linear(len(kernel_sizes) * num_filters, num_classes)
        self.dropout = nn.Dropout(0.5)

    def forward(self, x):
        x = self.embedding(x) # (batch_size, seq_len, embed_dim)
        x = x.unsqueeze(1) # (batch_size, 1, seq_len, embed_dim)
        x = [torch.relu(conv(x)).squeeze(3) for conv in self.convs] # [(batch_size, num_filters, seq_len -
kernel_size + 1)]
        x = [torch.max_pool1d(i, i.size(2)).squeeze(2) for i in x] # [(batch_size, num_filters)]
        x = torch.cat(x, 1) # (batch_size, num_filters * len(kernel_sizes))
        x = self.dropout(x)
        x = self.fc(x) # (batch_size, num_classes)
        return x
```

Step 9 Define the model training functions.

```
# Define the model training functions.
def train_model(model, train_loader, criterion, optimizer, device, num_epochs=10):
    model.to(device)
    model.train()

    for epoch in range(num_epochs):
        total_loss = 0
        for batch_X, batch_y in train_loader:
            batch_X, batch_y = batch_X.to(device), batch_y.to(device)
            optimizer.zero_grad()
            outputs = model(batch_X)
            loss = criterion(outputs, batch_y)
            loss.backward()
            optimizer.step()
            total_loss += loss.item()

    print(f'Train-Epoch [{epoch+1}/{num_epochs}], Loss: {total_loss/len(train_loader):.4f}')
```

Step 10 Define the test functions.

```
# Define the test model.
def test_model(model, test_loader, device):
    model.eval()
    correct = 0
    total = 0
    with torch.no_grad():
        for batch_X, batch_y in test_loader:
            batch_X, batch_y = batch_X.to(device), batch_y.to(device)
            outputs = model(batch_X)
            _, predicted = torch.max(outputs.data, 1)
            total += batch_y.size(0)
            correct += (predicted == batch_y).sum().item()

    print(f'Test Accuracy: {100 * correct / total:.2f}%')
```

Step 11 Define the sentiment prediction functions.

```
# Predict sentiments.
def predict_sentiment(text, model, vocab, max_len=200):
    text = preprocess_text(text)
    sequence = text_to_sequence(text, vocab)
    sequence = pad_sequence(sequence, max_len)
    sequence = torch.tensor(sequence, dtype=torch.long).unsqueeze(0).to(device)
    with torch.no_grad():
        output = model(sequence)
        _, predicted = torch.max(output, 1)
    return "pos" if predicted.item() == 1 else "neg"
```

Step 12 Train the model on the main program, and test the model accuracy using the test set, and save the model.

```
# Main program
if __name__ == "__main__":
```

```
# Download and load data.
data_dir = "data/imdb"
download_imdb(data_dir)
reviews, labels = load_imdb_data(data_dir)

# Preprocess data.
reviews = [preprocess_text(review) for review in reviews]
vocab = build_vocab(reviews)
sequences = [text_to_sequence(review, vocab) for review in reviews]

# Pad the sequences and split the dataset.
max_len = 200
X = [pad_sequence(seq, max_len) for seq in sequences]
X = torch.tensor(X, dtype=torch.long)
y = torch.tensor(labels, dtype=torch.long)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Create datasets and data loaders.
train_dataset = IMDBDataset(X_train, y_train)
test_dataset = IMDBDataset(X_test, y_test)
train_loader = DataLoader(train_dataset, batch_size=64, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=64, shuffle=False)

# Define the model, loss function, and optimizer.
vocab_size = len(vocab)
embed_dim = 100
num_classes = 2
model = TextCNN(vocab_size, embed_dim, num_classes)
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

# Obtain the device.

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Using device: {device}')

# Train the model.
train_model(model, train_loader, criterion, optimizer, device, num_epochs=10)

# Test the model.
test_model(model, test_loader, device)

# Save the model.
model_path = "textcnn_model.pth"
torch.save(model.state_dict(), model_path)
print(f"Model saved to {model_path}")
```

Output:

```
Using device: cpu
Train-Epoch [1/10], Loss: 0.6857
Train-Epoch [2/10], Loss: 0.5247
Train-Epoch [3/10], Loss: 0.4623
Train-Epoch [4/10], Loss: 0.4034
Train-Epoch [5/10], Loss: 0.3492
Train-Epoch [6/10], Loss: 0.3010
Train-Epoch [7/10], Loss: 0.2539
Train-Epoch [8/10], Loss: 0.2090
Train-Epoch [9/10], Loss: 0.1751
Train-Epoch [10/10], Loss: 0.1385
Test Accuracy: 84.70%
Model saved to textcnn_model.pth
```

Step 13 Test the model.

```
# Test two English texts.
test_texts = [
    "This movie is absolutely fantastic! The acting was superb and the plot was engaging.", # Positive
    "I hated this movie. It was boring and the characters were poorly developed." # Negative
]
for text in test_texts:
    sentiment = predict_sentiment(text, model, vocab)
    print(f"Text: {text}\nsentiment: {sentiment}\n")
```

Output:

```
text: This movie is absolutely fantastic! The acting was superb and the plot was engaging.
emotion: pos

text: I hated this movie. It was boring and the characters were poorly developed.
emotion: neg
```

1.6 Questions

- Why the loss does not decrease but increases during LeNet and ResNet training?
 - Can the accuracy be further improved if we continue to increase the number of training epochs during CIFAR-10 training of LeNet? If it can or cannot, why?
 - How can we further improve network accuracy without changing the network structure?
-

Huawei AI Certification Tutorials

HCIA-AI

Model Deployment and Application Lab Guide

ISSUE: 4.0



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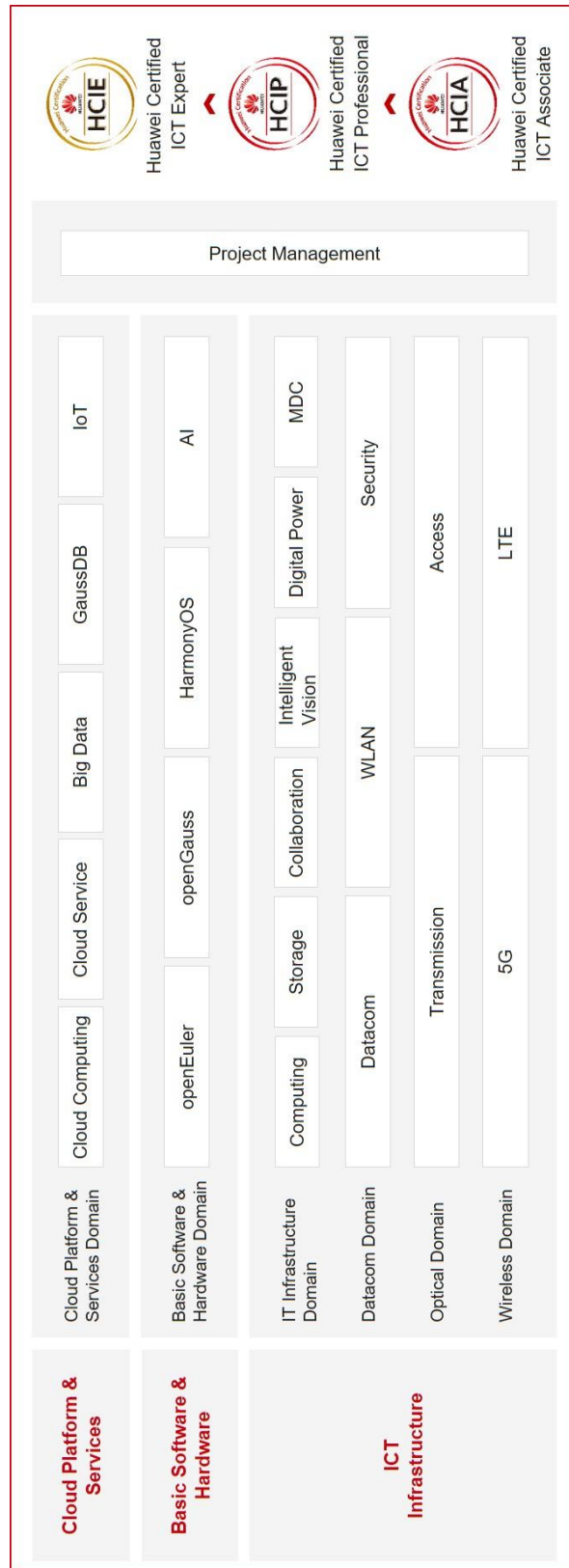
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About This Document

Overview

This document is intended for trainees who are preparing for the HCIA-AI exam as well as readers who are interested in the AI basics, Large Language Model (LLM) deployment, and API calling.

Description

This document consists of two experiments. Starting from CLI-based LLM deployment, it describes the operations such as CLI dialog, web-based LLM deployment, deployment service status query, and API dialog calling.

- Experiment 1 is CLI-based LLM deployment. You can communicate with LLM through the CLI by configuring basic parameters.
- Experiment 2 is web-based LLM deployment. After the service is successfully deployed, you can use simple code to call APIs to communicate with LLM.

Background Knowledge Required

This is a basic course for Huawei certification. To better understand this document, readers should:

- Have AI basics, Python programming capability, and computer knowledge.

Experiment Environment Overview

Device introduction

The following table lists the recommended device configurations for each HCIA-AI experiment environment.

Device, hardware, and software versions:

Device Name	Hardware Requirement	Software Version
PC	X86 CPU that supports the AVX instruction set.	Operating system: Windows 10 or later, or Ubuntu 22.04 or later Python version: Python 3.x

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1 CLI-based LLM Deployment

1.1 Experiment Introduction

1.1.1 About This Experiment

In this experiment, llama.cpp is used to deploy the model (project address: <https://github.com/gggerganov/llama.cpp>). The model file is the quantized DeepSeek-R1-Distill-Qwen-1.5B model (project image path: <https://huggingface.co/bartowski/DeepSeek-R1-Distill-Qwen-1.5B-GGUF>). llama.cpp is implemented using C++ and does not have any dependency, which greatly simplifies LLM deployment and is suitable for beginners. Requiring fewer resources for inference, the quantized DeepSeek-R1-Distill-Qwen-1.5B model delivers a high inference speed with the CPU on a laptop, lowering hardware requirements. It is the smallest inference model of the DeepSeek-R1 series and is capable of thinking. With a few simple steps, you can talk to LLM in the command line.

1.1.2 Experiment Objectives

- To learn how to use llama.cpp to deploy the LLM.
- To understand the common parameters for deploying the LLM.

1.2 Experiment Configuration

1.2.1 General Steps

- Download the llama.cpp executable file.
- Download the quantized DeepSeek-R1-Distill-Qwen-1.5B model file.
- Configure different running parameters to communicate with the LLM.

1.2.2 Procedure

Note: The procedure for downloading the code file and model file in a Windows operating system is different from that in Ubuntu. Perform operations based on the operating system. The deployment commands are the same.

1.2.2.1 Downloading the llama.cpp Executable File

1.2.2.1.1 Windows users

Step 1 Click the following link to download the compressed package:

https://certification-data.obs.cn-north-4.myhuaweicloud.com/CHS/HCIA-AI/V4.0/chapter5/llama_cpp.zip

Step 2 Decompress the package and open the folder, as shown in the following figure.

	test-tokenizer-0.exe	112 KB
	test-rope.exe	18 KB
	test-quantize-perf.exe	32 KB
	test-quantize-fns.exe	20 KB
	test-model-load-cancel.exe	12 KB
	test-log.exe	26 KB
	test-gguf.exe	54 KB
	test-chat-template.exe	917 KB
	test-c.exe	10 KB
	test-barrier.exe	19 KB
	test-backend-ops.exe	288 KB
	test-autorelease.exe	14 KB
	test-arg-parser.exe	1,065 KB
	rpc-server.exe	21 KB
	llava_shared.dll	253 KB

1.2.2.1.2 Ubuntu users

Step 1 Open a terminal window and run the following commands to download and decompress the file:

```
wget
https://certification-data.obs.cn-north-4.myhuaweicloud.com/CHS/HCIA-AI/V4.0/chapter5/llama_cpp_ubuntu.zip
unzip llama_cpp_ubuntu.zip
```

Step 2 Run the following command to open the folder and view the file:

```
cd build/bin && ll
```

```
[root@Atlas800T2 bin]# ll
total 159700
-rw-r--r--. 1 root root 1078 LICENSE
-rwxr-xr-x. 1 root root 3176480 llama-baby-llama
-rwxr-xr-x. 1 root root 3676328 llama-batched
-rwxr-xr-x. 1 root root 3676416 llama-batched-bench
-rwxr-xr-x. 1 root root 3257576 llama-bench
-rwxr-xr-x. 1 root root 935088 llama-bench-matmult
-rwxr-xr-x. 1 root root 3717080 llama-cli
-rwxr-xr-x. 1 root root 3127896 llama-convert-llama2c-to-ggml
-rwxr-xr-x. 1 root root 3699232 llama-cvector-generator
-rwxr-xr-x. 1 root root 3680752 llama-embedding
-rwxr-xr-x. 1 root root 3676576 llama-eval-callback
-rwxr-xr-x. 1 root root 3704368 llama-export-lora
-rwxr-xr-x. 1 root root 2782008 llama-gbnf-validator
-rwxr-xr-x. 1 root root 939744 llama-gguf
-rwxr-xr-x. 1 root root 1007888 llama-gguf-hash
-rwxr-xr-x. 1 root root 2787264 llama-gguf-split
-rwxr-xr-x. 1 root root 3681008 llama-gritlm
-rwxr-xr-x. 1 root root 3714704 llama-imatrix
-rwxr-xr-x. 1 root root 3699600 llama-infill
-rwxr-xr-x. 1 root root 3951344 llama-llava-cli
-rwxr-xr-x. 1 root root 3681032 llama-lookahead
-rwxr-xr-x. 1 root root 3701688 llama-lookup
-rwxr-xr-x. 1 root root 3697232 llama-lookup-create
-rwxr-xr-x. 1 root root 982432 llama-lookup-merge
-rwxr-xr-x. 1 root root 3702016 llama-lookup-stats
-rwxr-xr-x. 1 root root 3946696 llama-minicpmv-cli
-rwxr-xr-x. 1 root root 3685296 llama-parallel
-rwxr-xr-x. 1 root root 3676272 llama-passkey
-rwxr-xr-x. 1 root root 3769448 llama-perplexity
-rwxr-xr-x. 1 root root 934568 llama-q8dot
-rwxr-xr-x. 1 root root 3130824 llama-quantize
-rwxr-xr-x. 1 root root 2819624 llama-quantize-stats
-rwxr-xr-x. 1 root root 3694488 llama-retrieval
-rwxr-xr-x. 1 root root 3676496 llama-save-load-state
```

1.2.2.2 Downloading the DeepSeek-R1-Distill-Qwen-1.5B Model File

1.2.2.2.1 Windows users

Step 1 Click the following link to download the model file:

https://huggingface.co/bartowski/DeepSeek-R1-Distill-Qwen-1.5B-GGUF/resolve/main/DeepSeek-R1-Distill-Qwen-1.5B-Q6_K_L.gguf

Step 2 Move the file to the **llama_cpp** folder, as shown in the following figure.

 DeepSeek-R1-Distill-Qwen-1.5B-Q6_K_L.gguf	1,540,254 KB
 ggml.dll	64 KB
 ggml-base.dll	478 KB
 ggml-cpu.dll	414 KB
 ggml-rpc.dll	59 KB
 linenoise.cpp.txt	2 KB
 llama.cpp.txt	2 KB
 llama.dll	1,195 KB
 llama-batched.exe	1,061 KB
 llama-batched-bench.exe	1,060 KB
 llama-bench.exe	215 KB
 llama-cli.exe	1,128 KB
 llama-convert-llama2c-to-ggml.exe	125 KB
 llama-cvector-generator.exe	1,093 KB
 llama-embedding.exe	1,079 KB

1.2.2.2.2 Ubuntu users

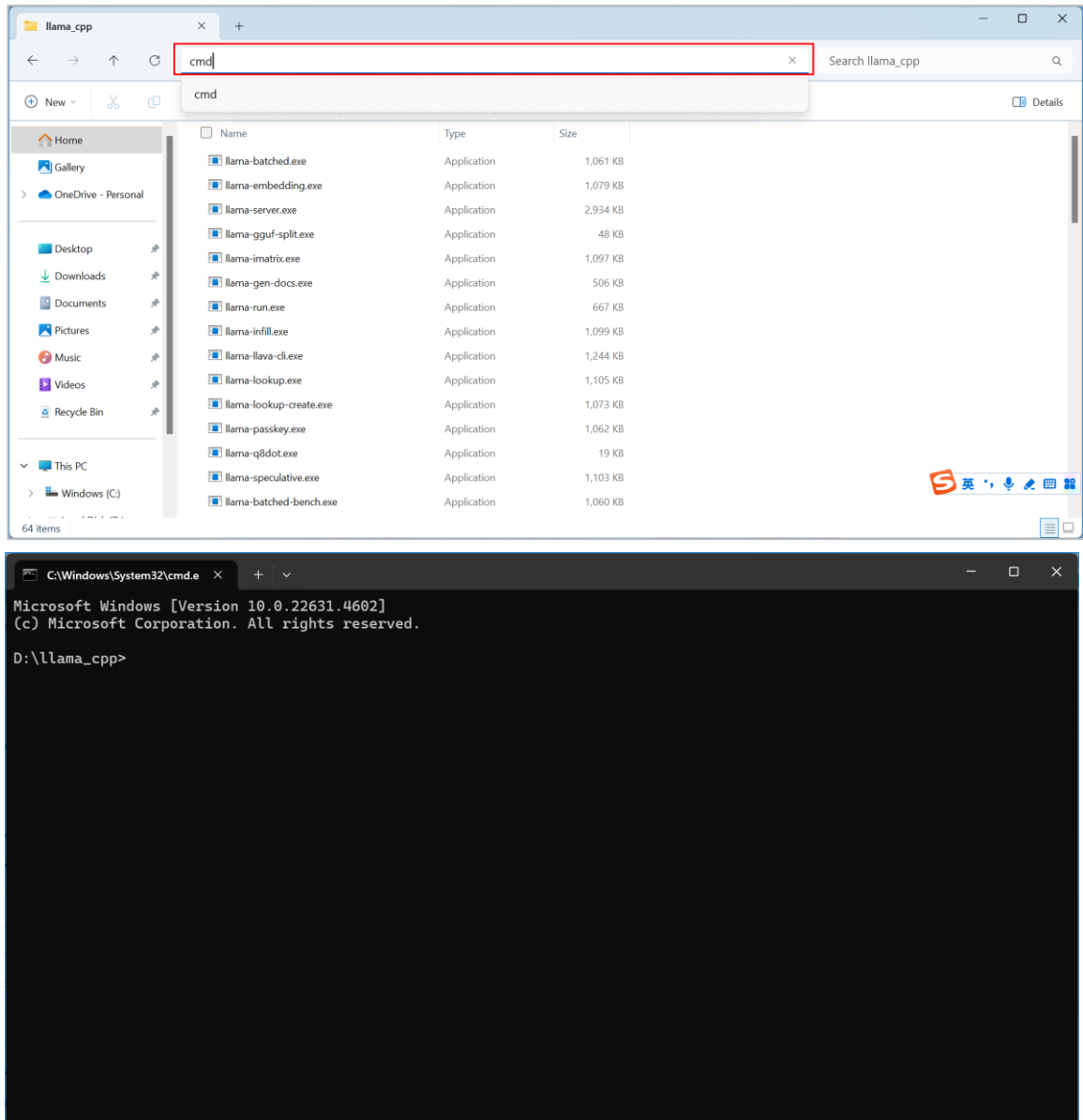
Step 1 Run the following command to download the model file to the code folder:

```
wget
https://huggingface.co/bartowski/DeepSeek-R1-Distill-Qwen-1.5B-GGUF/resolve/main/DeepSeek-R1-Distill-Qwen-1.5B-Q6_K_L.gguf
```

1.2.2.3 CLI-based Deployment

Step 1 Click the navigation bar above the folder, enter the following command, and press **Enter**. Ubuntu users can skip this step.

```
cmd
```



Step 2 Run the following command to perform inference for the model:

```
#Dialog mode
llama-cli.exe -m DeepSeek-R1-Distill-Qwen-1.5B-Q6_K_L.gguf --temp 0.6
```

Parameter description:

-m: model path

-cnv: conversation mode (Special tokens, prefixes, and suffixes are not printed, and the default chat template is used. By default, the dialog mode is enabled for the software package provided in this experiment. Adding parameters will lower the model's answer quality, so this parameter is not added to the CLI.)

--chat-template: chat template. If the template does not match the model, the model returns garbled characters or answers non-stop. For details, see https://github.com/gganganov/llama.cpp/wiki/Templates-supported-by-llama_chat_apply_template

--color: Different colors are used to distinguish user input from model output.

--temp: Adjusts the randomness of the output text. The default value is **0.8**. Set this parameter to **0.6**, which is recommended by DeepSeek.

--repeat-penalty: Controls repetition of the token sequence in the generated text. The default value is **1.0** (disabled).

--top-k: The next token is selected from the k candidates with the highest probability. The default value is **40**.

For more details about the parameters, click the following link:

<https://github.com/ggernanov/llama.cpp/tree/master/examples/main>

Step 3 Dialog with the model. For example, enter "who are you?" and obtain the result.

Step 4 The effects of different parameters are as follows (Text inside <think></think> is the model's thinking process):

```
llama-cli.exe --color -m DeepSeek-R1-Distill-Qwen-1.5B-Q6_K_L.gguf
```

```

C:\Windows\System32\cmd.exe x + v
sampler seed: 3565647506
sampler params:
  repeat_last_n = 64, repeat_penalty = 1.000, frequency_penalty = 0.000, presence_penalty = 0.000
  dry_multiplier = 0.000, dry_base = 1.750, dry_allowed_length = 2, dry_penalty_last_n = 4096
  top_k = 40, top_p = 0.950, min_p = 0.050, xtc_probability = 0.000, xtc_threshold = 0.100, typical_p = 1.000, tem
p = 0.800
  mirostat = 0, mirostat_lr = 0.100, mirostat_ent = 5.000
sampler chain: logits -> logit-bias -> penalties -> dry -> top-k -> typical -> top-p -> min-p -> xtc -> temp-ext -> dist
generate: n_ctx = 4096, n_batch = 2048, n_predict = -1, n_keep = 1

== Running in interactive mode. ==
- Press Ctrl+C to interject at any time.
- Press Return to return control to the AI.
- To return control without starting a new line, end your input with '/'.
- If you want to submit another line, end your input with '\'.
- Using default system message. To change it, set a different value via -p PROMPT or -f FILE argument.

You are a helpful assistant

> Who are you?
<think>

</think>

Greetings! I'm DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I'm at your service and would be d
elighted to assist you with any inquiries or tasks you may have.

> |

```


2 Web-based LLM Deployment

2.1 Experiment Introduction

2.1.1 About This Experiment

Different from exercise 1, llama.cpp is used to deploy the web service, and then APIs are called through code to dialog with the model. llama.cpp provides basic API services as well as OpenAI-compatible API services. You can select the calling mode as required. The llama.cpp code and model file are the same as those in exercise 1, and the code for calling APIs using Python is added.

2.1.2 Experiment Objectives

- To learn how to write Python code to call APIs
- To learn how to realize structured output of web return values
- To learn how to use llama.cpp to deploy web services.

2.2 Experiment Configuration

2.2.1 General Steps

- Use llama.cpp to deploy the web service.
- Use the library provided in Python to call APIs.
- Use the OpenAI library to call OpenAI-compatible APIs.

2.2.2 Procedure

Step 1 The llama.cpp code and model file are the same as those in experiment 1. Open the CLI and run the following command to deploy the web service:

```
llama-server.exe -m DeepSeek-R1-Distill-Qwen-1.5B-Q6_K_L.gguf -c 2048
```

Parameter description:

-m: model path

-c: length of the prompt text. The greater length, the more content the model can process. The default value is **4096**, and **0** indicates that the content is loaded from the model.

For more details about the parameters, click the following link:

<https://github.com/ggerganov/llama.cpp/tree/master/examples/server>

```
import requests
import json

def get_completion():
    response = requests.get(url='http://localhost:8080/health')
    return response

print(get_completion())

<Response [200]>
```

Step 4 Enter the following code to use the requests library provided by Python to call the original API. When entering prompts, specify whether it is the user's or the model's input. For details about other parameter configuration, click the following link: <https://github.com/gganganov/llama.cpp/tree/master/examples/server#post-completion-given-a-prompt-it-returns-the-predicted-completion>.

```
import requests
import json

def get_completion(prompt):
    headers = {'Content-Type': 'application/json'}
    data = {"prompt": prompt}
    response = requests.post(url='http://localhost:8080/completions', headers=headers,
data=json.dumps(data))
    return response.json()['content']

print(get_completion('<|User|>Hello<|Assistant|>'))
```

```
import requests
import json

def get_completion(prompt):
    headers = {'Content-Type': 'application/json'}
    data = {"prompt": prompt}
    response = requests.post(url='http://localhost:8080/completions', headers=headers, data=json.dumps(data))
    return response.json()['content']

print(get_completion('<| User |>Hello<| Assistant |>'))

<think>

</think>

Hello! How can I assist you today? 😊
```

Step 5 Enter the following code to use the requests library provided by Python to call the OpenAI-compatible API. This method is much more convenient. The final output is a list. You can use regular expressions to extract required content.

```
import requests
import json
def get_completion(prompt):
    headers = {'Content-Type': 'application/json','Authorization':'Bearer no-key'}
    data = {
"model": "gpt-3.5-turbo",
"messages": [
```

```
{
    "role": "user",
    "content": prompt
}
]
}

response = requests.post(url='http://localhost:8080/v1/chat/completions', headers=headers,
data=json.dumps(data))
return response.json()['choices']

print(get_completion('Hello'))
```

```
import requests
import json
def get_completion(prompt):
    headers = {'Content-Type': 'application/json', 'Authorization': 'Bearer no-key'}
    data = {
        "model": "gpt-3.5-turbo",
        "messages": [
            {
                "role": "user",
                "content": prompt
            }
        ]
    }
    response = requests.post(url='http://localhost:8080/v1/chat/completions', headers=headers, data=json.dumps(data))
    return response.json()['choices']

print(get_completion('Hello'))

[{'finish_reason': 'stop', 'index': 0, 'message': {'content': '<think>\n\n</think>\n\nHello! How can I assist you today? 😊', 'tool_calls': None, 'role': 'assistant'}}]
```

Step 6 Enter the following code to use the OpenAI library to call the OpenAI-compatible API. If you are prompted that the OpenAI library cannot be imported, run the following command to install the library. Using this method, you can easily obtain the required output and configure running parameters.

```
#Installation dependencies
pip install openai
```

```
import openai

client = openai.OpenAI(
    base_url="http://localhost:8080/v1", # Service request address for local deployment
    api_key = "sk-no-key-required"
)

completion = client.chat.completions.create(
    model="gpt-3.5-turbo",
    messages=[

        {"role": "user", "content": "Who are you?"}
    ]
)
```

```
print(completion)
print(completion.choices[0].message)
print(completion.choices[0].message.content)
```

```
import openai

client = openai.OpenAI(
    base_url="http://localhost:8080/v1", # Service request address for local deployment
    api_key = "sk-no-key-required"
)

completion = client.chat.completions.create(
    model="gpt-3.5-turbo",
    messages=[
        {"role": "user", "content": "Who are you?"}
    ]
)

print(completion)
print('-----')
print(completion.choices[0].message)
print('-----')
print(completion.choices[0].message.content)
```

ChatCompletion(id='chatcmpl-bTkivFy3IApSOWSBOTQ0ehPHtDZTDkoP', choices=[Choice(finish_reason='stop', index=0, logprobs=None, message=ChatCompletionMessage(content='<think>\n\n</think>\n\nGreetings! I\'m DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I\'m at your service and would be delighted to assist you with any inquiries or tasks you may have.', refusal=None, role='assistant', audio=None, function_call=None, tool_calls=None)), created=1740038205, model='gpt-3.5-turbo', object='chat.completion', service_tier=None, system_fingerprint='b4689-90e4dba4', usage=CompletionUsage(completion_tokens=44, prompt_tokens=7, total_tokens=51, completion_tokens_details=None, prompt_tokens_details=None), timings={'prompt_n': 5, 'prompt_ms': 81.719, 'prompt_per_token_ms': 16.343799999999998, 'prompt_per_second': 61.185281268737995, 'predicted_n': 44, 'predicted_ms': 1835.851, 'predicted_per_token_ms': 41.723886363637, 'predicted_per_second': 23.9670866535465})

ChatCompletionMessage(content='<think>\n\n</think>\n\nGreetings! I\'m DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I\'m at your service and would be delighted to assist you with any inquiries or tasks you may have.', refusal=None, role='assistant', audio=None, function_call=None, tool_calls=None)

<think>

</think>

Greetings! I\'m DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I\'m at your service and would be delighted to assist you with any inquiries or tasks you may have.

2.3 Question

When deploying the LLM, which parameters have the greatest impact on the model output?